# Unit 7 Overview: Rewriting Polynomial Expressions

## Unit Narrative

In this unit, students are introduced to factoring polynomials. Students have learned about and used the distributive property since Grade 3, applying it to multiply both numerical in elementary and, later, algebraic expressions in middle school. In their previous course (Grade 8 Pre-Algebra or Grade 7 Accelerated Mathematics), students also learned to factor algebraic expressions consisting of two terms by a common monomial factor. This unit will extend this knowledge deeper to focus on how to factor various types of polynomials.

In Section 1, students begin by exploring the concept of factoring by connecting it to prior learning about the distributive property. This includes introducing the concept of the greatest common monomial factor. In Lesson 2, students use algebra tiles to develop understanding of the concept of factoring a trinomial with a leading coefficient of one, connecting this to their prior knowledge of multiplying binomials. Lesson 3 guides students from the concrete use of algebra tiles to pictorial and abstract representations as understanding progresses. Students connect the concept of factoring to the pictorial area model and the abstract factoring by grouping. In Lesson 4, students will extend this learning by beginning to work with trinomials with a leading coefficient greater than one, first with the concrete algebra tiles and then into abstract representations. This section concludes in Lesson 5, where students apply what they have learned over the course of this section to factor a variety of trinomials using different methods.

In Section 2, students delve deeper into factoring polynomials, including those with more than three terms and special cases such as perfect squares and differences of squares. Lesson 6 focuses on factoring by grouping, including polynomial expressions with more than three terms. In Lesson 7, students learn to identify the existence of a perfect square trinomial and explore factoring such expressions. This section concludes in Lesson 8, where students are introduced to the concept of the difference of two squares and how to factor these expressions.

This unit lays the foundation for quadratic equations and functions that will be introduced in Units 8 and 9.

## Unit At-A-Glance

| | | | |
|---|---|---|---|
| Section 1: Factoring Trinomials | Benchmarks Addressed | Lesson 1 | Factoring and the Distributive Property |
| | **MA.912.AR.1.7** | Lesson 2 | Factoring a Trinomial With Algebra Tiles |
| | | Lesson 3 | Factoring a Trinomial With a Leading Coefficient of 1 |
| | | Lesson 4 | Factoring a Trinomial With a Leading Coefficient Greater Than 1 |
| | | Lesson 5 | Factoring Trinomials |
| Section 2: Factoring Special Cases | Benchmarks Addressed | Lesson 6 | Factoring by Grouping |
| | **MA.912.AR.1.7** | Lesson 7 | Factoring Perfect Square Trinomials |
| | | Lesson 8 | Factoring the Difference of Two Squares |

## Differentiation

| Universal Design for Learning™ © CAST, 2008 | |
|---|---|
| **Engagement**<br>*The "WHY" of learning* | • Create competition and game-like experiences where appropriate. For example, in 7.1.2 Exploration in Lesson 1, consider turning Question 10 into a challenge for students to find all the possible equivalent expressions they can. This can be a game or challenge to determine which student or team can find the most equivalent expressions.<br>• With 7.3.2 Collaboration in Lesson 3, consider creating a debate-style activity out of Question 7. Instead of having students simply write their responses and discuss with a partner, structure a class-wide debate where students use their reasoning and rationale to make their mathematical case. |
| **Representation**<br>*The "WHAT" of learning* | • Use physical manipulatives within the lessons where possible. For example, in 7.2.2 Exploration in Lesson 2, consider using physical algebra tiles. If technology exists, consider showing the use of algebra tiles using a document camera or providing access to virtual algebra tiles for students to use throughout the lessons as needed.<br>• As students learn different methods for factoring, make note of which methods students connect with most so their understanding of these methods can be leveraged when providing differentiated support to students throughout the unit. |
| **Action & Expression**<br>*The "HOW" of learning* | • With 7.4.1 Warm-Up in Lesson 4, consider the use of a tool like FlipGrid to allow students to respond to the discussion questions in this component rather than writing a response.<br>• Consider turning the card sort in 7.5.3 Activity in Lesson 5 digital by using a program such as Desmos to make the card sort activity more interactive and in a digital platform that your students may feel more engaged with. |

## ELL Information

- Connect factoring polynomials to the models used to learn multiplying polynomials in Unit 2. Make sure that ELL students understand that the area model and the distributive property will both work for multiplying polynomials. It may help to highlight each term with a different color and be consistent in using the same colors for each term as you work through multiplying polynomial problems using both methods.
- Build understanding that factoring is the inverse of multiplying. Have students point to the corresponding term in the example given in the book as you discuss each bullet point. It may help to do this for several additional examples as well.
- When guiding students through their first factoring problems, clearly write out each step and pause to allow students to solidify their understanding of where each factor comes from as they process and ask questions. It may help to have an anchor chart with examples for ELL students to refer to whenever needed.
- Students should begin making a list of the different forms and special cases for quadratic equations in their notebooks. The teacher should also post a big version of this list in the classroom. ELL students will benefit from having this list in both places so they are able to follow along with the teacher. By the end of this unit, the list can include: standard form of a quadratic equation, perfect square trinomial, and difference of two squares.

All glossary terms are available in multiple languages including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Area model
- Quadratic expression (or quadratic equation)

## Additional Differentiated Support

| Scaffolding Resources | On-Ramp |
|---|---|
| • Continue the use of algebra tiles beyond the specific lessons where called to do so to support students who need additional support.<br><br>• Refer back to the area model to remind students that factoring out a GCF is the inverse of distributing a single term. Have students work forward and backward, using the distributive property and then factoring on the same problem to solidify this understanding. | Use the videos and practice problems in the On-Ramp to Algebra 1 personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.<br><br>• Distributive Property<br>• Writing Expressions (Distributive Property)<br>• Factoring Linear Expressions<br>• Determining the GCF between Integers<br>• Identifying Coefficients<br>• Identifying Variables<br>• Writing Expressions (Combining Like Terms) |

# Section Coherence

## Section 1: Factoring Trinomials (Lessons 1-5)

| Benchmarks Addressed | Learning Targets | |
|---|---|---|
| **MA.912.AR.1.7** Rewrite a polynomial expression as a product of polynomials over the real-number system.<br>• *Clarification 1: Within the Algebra 1 course, polynomial expressions are limited to* $4$ *or fewer terms with integer coefficients.* | Lesson 1 | • I can rewrite a polynomial using the distributive property by factoring out a common factor.<br>• I can determine the greatest common monomial factor. |
| | Lesson 2 | • I can use algebra tiles to factor a trinomial with a leading coefficient of one. |
| | Lesson 3 | • I can factor a trinomial with a leading coefficient of 1. |
| | Lesson 4 | • I can use algebra tiles to factor a trinomial with a leading coefficient greater than 1.<br>• I can factor a trinomial with a leading coefficient greater than 1. |
| | Lesson 5 | • I can factor trinomials. |

## Section 2: Factoring Special Cases (Lessons 6-8)

| Benchmarks Addressed | Learning Targets | |
|---|---|---|
| **MA.912.AR.1.7** Rewrite a polynomial expression as a product of polynomials over the real-number system.<br>• *Clarification 1: Within the Algebra 1 course, polynomial expressions are limited to* $4$ *or fewer terms with integer coefficients.* | Lesson 6 | • I can factor a polynomial by grouping. |
| | Lesson 7 | • I can factor perfect square trinomials. |
| | Lesson 8 | • I can factor a difference of two squares.<br>• I can rewrite a polynomial expression as a product of polynomials by factoring. |

## Vertical Alignment

| Grade 8 Unit 12<br>Equivalent Algebraic Expressions | → | **Algebra 1 Unit 7<br>Factoring Polynomials** | → | Algebra 2 Unit 2<br>Polynomials and the Complex Number System |
|---|---|---|---|---|

| Prior Course(s) | Course Level | Future Course(s) |
|---|---|---|
| **MA.8.AR.1.3** Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | **MA.912.AR.1.7** Rewrite a polynomial expression as a product of polynomials over the real-number system. | **MA.912.AR.1.8** Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

# 7.1 Factoring and the Distributive Property

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Targets | • I can rewrite a polynomial using the distributive property by factoring out a common factor.<br>• I can determine the greatest common monomial factor. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

| | |
|---|---|
| Essential Questions | • How can you rewrite a polynomial using the distributive property?<br>• How can you determine the greatest common monomial factor? |
| Lesson Narrative | In this lesson, students extend their prior learning about the distributive property to factoring polynomial expressions by a common monomial factor. Students will be introduced to the concept of common monomial factors and the greatest common monomial factor. They will practice factoring polynomials by a common monomial factor with the distributive property. The lesson focuses on rewriting multiple equivalent expressions for a given polynomial. |

| | | | | |
|---|---|---|---|---|
| Component Summary | 7.1.1 Warm-Up (~5 min.) | Electric Motorcycle Engineer career video *(SE p. 357)* | 7.1.4 Guided Practice (10 min.) | Rewrite polynomials using common factors and the distributive property *(SE p. 361)* |
| | 7.1.2 Exploration (10-15 min.) | Explore connections between factors and products using the distributive property *(SE p. 358)* | 7.1.5 Your Turn! (5 min.) | Practice rewriting polynomials using common factors and the distributive property *(SE p. 361)* |
| | 7.1.3 Guided Instruction (10 min.) | Determine the greatest common monomial factor in polynomial expressions *(SE p. 360)* | 7.1.6 Wrap-Up (~5 min.) | Ticket Out the Door rewriting an expression using common monomial factors *(SE p. 362)* |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | Key Terms:<br>• Equivalent<br><br>Additional Glossary: N/A | • (Optional) Warm-Up video | • N/A |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may make errors when identifying the greatest common monomial factor.</li><li>Students may make basic divisibility rule errors when looking for common factors.</li><li>Students may try to factor out a variable that not all terms share.</li></ul></td><td><ul><li>Students have already been exposed to factoring using the distributive property in either Grade 8 Prealgebra or Grade 7 Accelerated Mathematics, but if students are placed into Algebra 1 without going through the $8^{th}$-grade benchmarks, then factoring using the distributive property will be new to them.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect<br>In 7.1.3 Guided Instruction, students receive formal instruction on common monomial factors. This concept may need additional review of divisibility rules. Consider leveraging this routine to help share thinking out loud while walking students through the process.</td><td>MA.K12.MTR.4.1: Discussion and Reflection<br>7.1.2 Exploration has students work together to explore the distributive property in factoring. Students will work with a partner to engage in structured discussions and reflections that lead to the concept of determining common monomial factors in the following activity.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">7.1.2 Exploration has students work together in a collaborative setting to explore the distributive property in factoring. Consider pairing students together strategically based on their learning styles and prior formative data.<br><br>Consider creating an anchor chart for the process of determining the greatest common monomial factor of monomial terms. This anchor chart can capture the processes described in 7.1.3 Guided Instruction, with specific reference to the table in Question 6 for examples.<br><br>The 7.1.6 Wrap-Up for this lesson provides a formative assessment point to determine students' mastery with common monomial factors prior to the next lessons.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.1.1 Warm-Up: Electric Motorcycle Engineer

***Component Overview**: Students will engage in a review of a quote from an electric motorcycle engineer. This activity is designed to be done independently.*

## 7.1.1 Warm-Up: Electric Motorcycle Engineer

1. Marc Fenigstein says, "Open-ended problems are the problems students will face in the real world … Forcing students to learn how to function in ambiguity and find their own answers … and that there isn't necessarily any one right answer, but there might be better or worse answers … I think that is really valuable. Don't be afraid to find your own path … You're not going to get it right on the first try, but you can make almost any job your dream job by looking for opportunities to do more, to go beyond …"
   a. What stands out to you the most in this statement?

      *Answers will vary. What stands out most to me is how open-ended problems are what we will face in the real world. What stands out to me is the idea that you can make any job your dream job.*
   b. Why does that part of the statement stand out to you?

      *Answers will vary. It stands out to me because those are the problems I avoid in class, but I should embrace them to prepare for the real-world. It stands out to me because it helps me see that I can take any job I'm in and make it amazing.*

*Display and show the video to students prior to their responding to the prompts.*

*Consider highlighting any exemplar responses that connect to MTR Standards. For example, highlight responses like "I like how in this class we talk more about how we got to the right answer than the actual answer itself," as this connects to multiple representations (MA.K12.MTR.2.1) and flexibility in fluency (MA.K12.MTR.3.1).*

# 7.1.2 Exploration: The Distributive Property in Factoring

***Component Overview****: Students will work together with a partner or small group to explore the distributive property in factoring.*

## 7.1.2 Exploration: The Distributive Property in Factoring

1. Multiply $9(10+6)$ using the distributive property.
   $(9)(10)=90$
   $(9)(6)=54$
   $90+54=144$
2. Discuss with your partner how $9(10+6)$ shows that $9$ and $16$ are factors of $144$.
   *Answers will vary. The product of $9$ and $16$ is $144$. Therefore, those two would be factors of $144$.*
3. Discuss with your partner how multiplying $8(287)$ can be done more easily using the distributive property. Explain or show your work.
   *Answers will vary. Taking the value of $287$ and breaking it into the sum of other terms can make multiplying easier. $8(287)$ can be rewritten as $8(280+7)$ to become $2{,}240+56$, which is $2{,}296$.*
4. Rewrite the expression $-7(2x-5)$ using the distributive property.
   $(-7)(2x)=-14x$
   $(-7)(-5)=35$
   $-14x+35$
5. Complete the table.

| Expression | Factors | Product |
|---|---|---|
| $9(10+6)$ | $9$ and $16$ | $144$ |
| $-7(2x-5)$ | $-7$ and $2x-5$ | $-14x+35$ |
| $(8-4x)10$ | $8-4x$ and $10$ | $80-40x$ |
| $-12(7-3x)$ | $-12$ and $7-3x$ | $-84+36x$ |
| $2(6x^2+8x)$ | $2$ and $6x^2+8x$ | $12x^2+16x$ |
| $4x(3x+4)$ | $4x$ and $3x+4$ | $12x^2+16x$ |
| $(6x+8)\cdot 2x$ | $6x+8$ and $2x$ | $12x^2+16x$ |

*This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.*

*Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.1.3 Guided Instruction.*

*Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.*

*How did you break the number 287 up to apply the distributive property more easily?*

*Why do you think it's possible that you can break it into different sums and get the same answer when you distribute?*

6. What do you notice about the product of each of the expressions $2(6x^2 + 8x)$, $4x(3x + 4)$, and $(6x + 8) \cdot 2x$?
*All of the products are the same.*

Expressions are ***equivalent*** if they have the same value no matter what number is substituted in for each of the variables.

7. The expression $14x^2 + 28x^3$ is the product of each of the following expressions.
$2(7x^2 + 14x^3)$ $\quad x^2(14 + 28x)$ $\quad 7x^2(2 + 4x)$

List the factors for each expression.

| Factors of $2(7x^2 + 14x^3)$ | | Factors of $x^2(14 + 28x)$ | | Factors of $7x^2(2 + 4x)$ | |
|---|---|---|---|---|---|
| $2$ | $7x^2 + 14x^3$ | $x^2$ | $14 + 28x$ | $7x^2$ | $2 + 4x$ |

8. Write two more expressions that are equivalent to $14x^2 + 28x^3$.
*Answers will vary.* $14(x^2 + 2x^3)$ *and* $2x(7x + 14x^2)$. $14x(x + 2x^2)$ *and* $7(2x^2 + 4x^3)$.

9. Compare your expressions with your partner's expression. Write the expressions that you and your partner do not have in common.
*Answers will vary.*

10. Complete the table for the factors of $14x^2 + 28x^3$ using two other expressions you created.

| Monomial Factor | Binomial Factor |
|---|---|
| $2$ | $(7x^2 + 14x^3)$ |
| $x^2$ | $(14 + 28x)$ |
| $7x^2$ | $(2 + 4x)$ |
| $2x$ | $(7x + 14x^2)$ |
| $14x^2$ | $(1 + 2x)$ |

*Consider keeping a copy of this table on a note card with the correct answers. This will help when you are circulating the room to quickly check students' answers to see who may need additional support reviewing these concepts.*

*Consider creating a visual representation of the word "equivalent." This can include multiple examples with justification for why those examples meet the conditions of equivalency.*

*Consider having students compare their work with an additional student partner pair to look for additional similarities and differences with the equivalent expressions in question 9.*

# 7.1.3 Guided Instruction: Common Monomial Factors

***Component Overview:*** *In this Guided Instruction, students will receive formal instruction on determining common monomial factors, including the greatest common monomial factor, for a given expression.*

## 7.1.3 Guided Instruction: Common Monomial Factors

The table in the exploration shows there are several different ways to rewrite the expression $14x^2 + 28x^3$.

1. What monomial factor is the greatest common monomial factor of $14x^2 + 28x^3$?
   $14x^2$
2. Discuss with your partner how to determine if a monomial factor is the greatest common monomial factor of a given polynomial. Summarize your discussion.
   *When the polynomial being factored can no longer be factored further with additional monomial factors, you have found the greatest common monomial factor.*
3. Determine the greatest common factor for $24, -36$, and $-54$.
   $6$
4. Determine the greatest common factor for $x^7$, $x^{10}$, and $x^{13}$.
   $x^7$
5. Determine the greatest common factor for $x^3y^3$, $x^2y^8$, and $xy^4$.
   $xy^3$
6. Work with your partner to complete the table. If an expression does not have one of the greatest common factors listed, write N/A.

| Expression | Greatest Common Numeric Factor | Greatest Common Variable Factor | Greatest Common Monomial Factor |
|---|---|---|---|
| $25w^3 - 40w^2 + 60w$ | $5$ | $w$ | $5w$ |
| $81g + 108g^2 - 12g^4$ | $3$ | $g$ | $3g$ |
| $7m^3n^2 + 56m^2n^3 - 21mn$ | $7$ | $mn$ | $7mn$ |
| $12p^2q + 16q^2 - 64p$ | $4$ | $N/A$ | $4$ |

*Students learned to find the greatest common factor of numerical expressions in Grade 6. Guide students through the process of finding common factors and the GCF of monomials in this component.*

*What is the difference between a common factor and the greatest common variable factor?*

*How can you tell if the common factor you found is the greatest common variable factor?*

*If time permits, consider turning question 6 into a matching activity. You can do this by taking the answers and placing them on index cards. Distribute the cards to all students and have them go around the room to find their correct match.*

# 7.1.4 Guided Practice: Factoring With the Distributive Property

***Component Overview***: *Students will engage in structured practice with factoring with the distributive property. Students should work independently or with their partner to apply what they learned in the prior activities.*

## 7.1.4 Guided Practice: Factoring with the Distributive Property

1. Which of the following expressions are equivalent to $8a^3 - 98a$? Select all that apply.

   - ☐ $a(8a^2 - 98)$
   - ☐ $2(4a^3 - 49a)$
   - ☐ $2a(4a^2 - 49)$
   - ☐ $2a(4a^3 - 49a)$
   - ☐ $4a(2a^2 - 49a)$
   - ☐ $8a^3(2 - 98a)$

2. Write three equivalent expressions for $18x^3 - 9x^2 - 15x$. One of the expressions should include the greatest common monomial factor.

   $3x(6x^2 - 3x - 5)$

   $x(18x^2 - 9x - 15)$

   $3(6x^3 - 3x^2 - 5x)$

3. Write an equivalent expression for $2c^3d^3 + 14c^2d + 18cd$ that uses the greatest common monomial factor.

   $2cd(c^2d^2 + 7c + 9)$

*Use observational data from the 7.1.2 Exploration and 7.1.3 Guided Instruction to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in prior components prior to releasing students to work on questions 2 and 3.*

*Think-Pair-Share*
*Consider leveraging a Think-Pair-Share strategy for this component. Have students try to complete these questions on their own and then work with a partner to compare their answers to come to consensus on each question prior to reviewing the component as a whole class.*

*For a challenge, have students compete to see who can come up with the most equivalent expressions for question 2. This can help students build up their factoring skills.*

# 7.1.5 Your Turn!

***Component Overview***: *Students will engage in an independent practice opportunity writing equivalent expressions using factors and the distributive property.*

## 7.1.5 Your Turn!

1. Write three equivalent expressions for $72b^2k^3 - 64b^5k^2 + 32b^8k^6$. One of the expressions should include the greatest common monomial factor.
   $8b^2k^2(9k - 8b^3 + 4b^6k^4)$
   $b(72bk^3 - 64b^4k^2 + 32b^7k^6)$
   $2(36b^2k^3 - 32b^5k^2 + 16b^8k^6)$

2. Which expressions are equivalent to $48r^3 + 18r^2$? Select all that apply.
   - ☐ $3(16r^3 + 6r^2)$
   - ☐ $6r(8r^2 + 3r)$
   - ☐ $2r^2(24r + 9)$
   - ☐ $4(12r^3 + 4r^2)$
   - ☐ $6r^2(8r + 3)$

*For this component, consider working with small groups of students who may have been struggling throughout this lesson.*

*Consider having students leverage an agree or disagree protocol. Display each expression and have students decide whether that expression is equivalent, and then have students vote on whether they agree or disagree with the answer before turning and talking with a partner to discuss their rationale.*

# 7.1.6 Wrap-Up: Ticket Out the Door

***Component Overview***: *Students will complete a Ticket Out the Door to demonstrate their levels of understanding of common monomial factors.*

## 7.1.6 Wrap-Up: Ticket Out the Door

1. Write three equivalent expressions for $12a^3b^5 - 16a^2b^2 + 20a^3b$. One of the expressions should include the greatest common monomial factor.

   $4a^2b(3ab^4 - 4b + 5a)$
   $2(6a^3b^5 - 8a^2b^2 + 10a^3b)$
   $a(12a^2b^5 - 16ab^2 + 20a^2b)$

*Student performance on this component can be used to determine what additional support, if any, may be needed in future lessons. Students may understand how to find common factors but miss the greatest common factor. Students may also factor incorrectly. Use responses from this Wrap-Up to inform differentiation in Lesson 2.*

# 7.2 Factoring a Trinomial With Algebra Tiles

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Target | • I can use algebra tiles to factor a trinomial with a leading coefficient of 1. |

| Benchmark Coherence | Building On | Working Toward |
|---|---|---|
| | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

| | |
|---|---|
| Essential Questions | • How can algebra tiles be used to show factoring a trinomial with a leading coefficient of one?<br>• What is the relationship between multiplication and factoring polynomials? |
| Lesson Narrative | In this lesson, students use algebra tiles to explore factoring trinomials. Students will connect multiplying binomials with algebra tiles to using that information to factor trinomial expressions. They will then spend time practicing the process of factoring trinomials with a leading coefficient of one. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 7.2.1 Warm-Up (~5 min.) | Multiply binomials using algebra tiles *(SE p. 363)* | 7.2.3 Guided Practice (10 min.) | Practice factoring trinomials using models *(SE p. 368)* |
| | 7.2.2 Exploration (20 min.) | Explore factoring trinomials using algebra tiles *(SE p. 363)* | 7.2.4 Your Turn! (5 min.) | Independent practice factoring trinomials using models *(SE p. 368)* |
| | 7.2.5 Wrap-Up (~5 min.) | Reflect on understanding and confidence factoring trinomials *(SE p. 369)* | | |

| Resources | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| | Key Terms: N/A<br><br>Additional Terms:<br>• Factoring | • (Optional) Algebra tiles | • N/A |

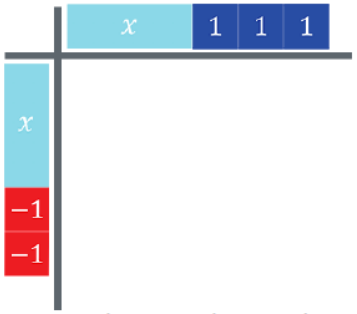
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may make errors with the signs of the algebra tiles.</li><li>Students may reverse the sum and product of the trinomial factoring process.</li></ul></td><td><ul><li>The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.</li><li>Using algebra tiles to model is a powerful tool to use in building students’ understanding of this concept. If you are unfamiliar with using these models, be sure to watch the video for this lesson to see how it unfolds in this structure.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns<br>7.2.2 Exploration is a collaborative exploration activity where students work together to model factoring with algebra tiles. During this activity, consider leveraging this routine to have students take turns with the tiles and steps within the process.</td><td>MA.K12.MTR.2.1: Multiple Representations<br>7.2.2 Exploration has students working with algebra tiles to model factoring. In 7.2.3 Guided Practice, students shift to factoring algebraically without the algebra tiles. Throughout this lesson, students will engage with factoring trinomials using multiple methods and representations. Guide students from concrete to pictorial to abstract representations as understanding progresses.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 7.2.2 Exploration activity, consider providing students with actual algebra tile manipulatives, paper cutouts of the algebra tiles, or access to virtual algebra tiles for this lesson.<br><br>7.2.4 Your Turn! provides students with an independent practice opportunity factoring trinomials where the leading coefficient is one. Student performance on this activity can provide additional data on whether students may need additional support in future lessons.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.2.1 Warm-Up: Multiplying Binomials

***Component Overview****: Students will practice multiplying binomials using algebra tiles. For this activity, students work individually to recall multiplying binomials using algebra tiles from Unit 2 Lesson 6.*

## 7.2.1 Warm-Up: Multiplying Binomials

1. An algebra tile mat is shown with two binomial factors.




   a. What are the two binomials represented?
   $(x+3)$ *and* $(x-2)$

   b. What is the product of the two binomials?
   $x^2+x-6$

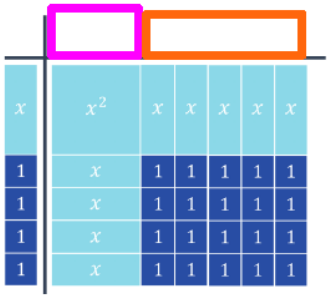
*Consider spending time reviewing the process and answers to this component. Use this time to ensure that students are familiar with how to read and use algebra tiles. Connect the prior lesson's focus on factors and products to identifying where the factors and products are represented in this model.*

# 7.2.2 Exploration: Model Factoring With Algebra Tiles

***Component Overview***: *Students will explore how to model factoring using algebra tiles. This activity is designed to be done with a partner.*

## 7.2.2 Exploration: Model Factoring with Algebra Tiles

1. An algebra tile mat is shown, where the product is the trinomial $x^2+9x+20$ and one of its factors is $(x+4)$.




Diego and Maria Jose both justified how they determined what belonged in the **pink** box in the following ways.

| Diego | Maria Jose |
|---|---|
| I found $\frac{x^2}{x}$ because $x^2$ is the product of $x$ and something else. | I asked myself what times $x$ would give me $x^2$. |

a. Discuss with your partner both justifications.
*Answers will vary.*

b. Use one of the methods to complete the binomial for the top bar of the algebra tile mat.

| | Drawing | Algebraic Representation |
|---|---|---|
| **Pink Box** | $x$ | $x$ |
| **Orange Box** | 1 1 1 1 1 | 5 |
| **Binomial Factor** | $x+5$ | |

*This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component.*

*Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.2.3 Guided Practice.*

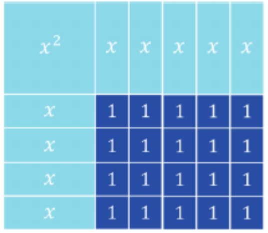
*Think-Pair-Share*
*Utilize this routine to give equitable opportunity for student thinking and learning. Students will be exploring in groups, so the Think-Pair-Share routine ensures that all students are engaged and not relying on one student to do the critical thinking.*

*Will Diego and Maria Jose get the same result based on what they have described? Why or why not?*

# 7.2.2 Exploration: Model Factoring With Algebra Tiles (continued)

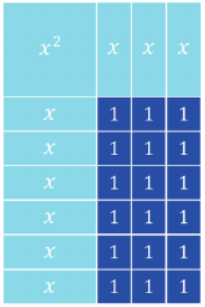
2. Discuss with your partner how to determine the two binomial factors when only given the product. Summarize your discussion.

*Answers will vary. Find the dimensions of the rectangle.*




The two binomials are the factors of the trinomial $x^2 + 9x + 20$. The process of finding the two binomials is called **factoring**.

3. The algebra tiles shown represent the trinomial $x^2 + 9x + 18$.




   a. Discuss with your partner what the similarities and differences are between the algebra tiles that represent $x^2 + 9x + 20$ and $x^2 + 9x + 18$. Summarize your discussion.

| | |
|---|---|
| **Similarities** | *Answers will vary. Both of them are complete without any gaps in tiles. Both of them have one $x^2$ tile.* |
| **Differences** | *Answers will vary. The first model is wider horizontally and the second one is taller. One model has 18 unit tiles while the other has 20.* |

   b. What are the two binomials that are factors of $x^2 + 9x + 18$?
   $(x + 3)$ *and* $(x + 6)$

*Revisit this portion when reviewing the activity and highlight exemplar student work or responses. Consider charting up their responses and seeing which other students may agree or disagree with their rationale.*

*After students have shared out their thinking, provide a structured Think Aloud that helps students connect the product mat to the missing algebra tiles that represent the factors.*

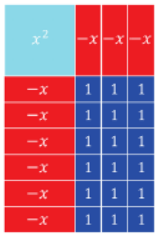
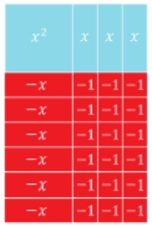
*Connect this process and thinking to the narrative language about factoring on this page.*

*Encourage connections between concrete observation and abstract representation.*

- *What do you notice about the algebra tiles?*
- *How might that be represented using algebra?*
- *What do you notice about the expression? How might that be represented or modeled using algebra tiles?*

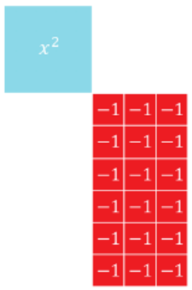
*MA.K12.MTR.1.1: Participate in Effortful Learning*
*Consider having students share out their approaches. Students will benefit from hearing a variety of strategies from their peers.*

4. The table shows the algebra tile representation for two trinomials. Work with your partner to determine the two binomial factors.

| Trinomial | $x^2-9x+18$ | $x^2-3x-18$ |
|---|---|---|
| **Algebra Tiles** |  |  |
| **Binomial Factors** | $(x-3)$ *and* $(x-6)$ | $(x-6)$ *and* $(x+3)$ |

One common attribute of all of the algebra tile representations is that they are all rectangles.

5. Consider the following algebra tile representation.




a. Complete the algebra tile representation of a trinomial by drawing and labeling $x$ or $-x$ tiles.
*Three positive $x$ tiles across the top and six negative $x$ tiles along the side. Or: Three negative $x$ tiles across the top and six positive $x$ tiles along the side.*

b. What is the trinomial that is represented by the completed drawing?
$x^2-3x-18$, or alternatively $x^2+3x-18$

c. What are the binomial factors of the trinomial?
$(x-6)(x+3)$ or alternatively, $(x+6)(x-3)$

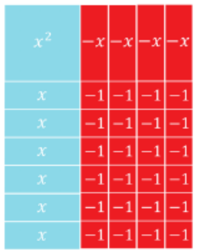
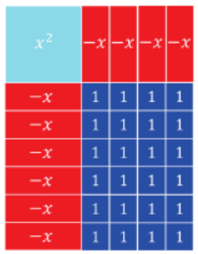
*For question 4, it can be beneficial to have partner pairs compare their work with another partner pair or group.*

*Physical manipulatives can be very helpful with helping students think through question 5. If you don't have actual algebra tiles available, consider creating cutouts using card stock or different colored paper to represent the algebra tiles or providing access to virtual algebra tiles. For students who responded better to hands-on or visual stimuli, treating this like a puzzle can help them complete it and then connect it to the abstract concepts being explored.*

*Provide space for productive struggle. The Exploration builds intentionally for explicit connections between $x$-tiles and their values (positive or negative).*

# 7.2.2 Exploration: Model Factoring With Algebra Tiles (continued)

6. The algebra tile representation of two different trinomials is shown.
   a. Complete the table.

| Algebra Tiles | Trinomial | Factors |
|---|---|---|
|  | $x^2 + 2x - 24$ | $(x-4)(x+6)$ |
|  | $x^2 - 10x + 24$ | $(x-6)(x-4)$ |

Each of the trinomials in the table has either twenty-four unit squares, $^{+}1$, or twenty-four negative unit squares, $^{-}1$. Also, each trinomial has some combination of four and six $^{+}x$ or $^{-}x$ tiles.

b. Discuss with your partner how the $x$-term of each trinomial relates to the four and six $^{+}x$ or $^{-}x$ tiles.
*Answers will vary. The $x$ term is the sum of the four and six $^{+}x$ or $^{-}x$ tiles.*

c. Discuss with your partner how the constant term, $24$ or $-24$, of each trinomial relates to the four and six $^{+}x$ or $^{-}x$ tiles.
*Answers will vary. If the constant is positive, the $x$ tiles would all need to be negative (or all positive). If the constant is negative, one set of the $x$ tiles would need to be positive and the other negative.*

*Use informal observation up to this point to determine whether additional guidance may be needed prior to completing the Exploration. This could mean pulling a small group or teacher facilitation with the whole group to ensure understanding prior to releasing students to complete the remainder of the Exploration with their partners.*

*Use inquiry for synthesis when reviewing this activity.*
- *What would it mean when the signs of the x-terms are different?*
- *How is the trinomial impacted when the signs of the x-terms are different?*

*When reviewing the answer to part a, consider allowing different students to share the trinomial and each of the binomial factors. Allow them to engage in an agree/disagree protocol where they are polled on whether they agree with the answer or disagree.*

*Leverage parts b and c from question 6 as a discussion point when reviewing as a whole group. The concepts developed here in these two questions are important for a deeper conceptual understanding.*

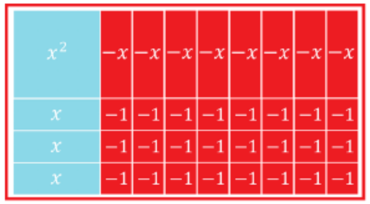
# 7.2.2 Exploration: Model Factoring With Algebra Tiles (continued)

7. Consider the trinomial $x^2 - 5x - 24$.

   a. Work with your partner to determine how many $^{+}x$ or $^{-}x$ tiles are needed for $x^2 - 5x - 24$.

| $^{+}x$ tiles | $^{-}x$ tiles |
|---|---|
| 3 | 8 |

   b. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 5x - 2$




   c. Use your algebra tiles diagram to choose the number of $^{+}x$ or $^{-}x$ tiles that match your diagram. Then, fill in each blank with the appropriate values.

| | $x$ | $-8$ |
|---|---|---|
| $x$ | $x^2$ | o $1x$ o $-1x$ o $6x$ o $-6x$<br>o $2x$ o $-2x$ o $8x$ o $-8x$<br>o $3x$ o $-3x$ o $12x$ o $-12x$<br>o $4x$ o $-4x$ o $24x$ o $-24x$ |
| $+3$ | o $1x$ o $-1x$ o $6x$ o $-6x$<br>o $2x$ o $-2x$ o $8x$ o $-8x$<br>o $3x$ o $-3x$ o $12x$ o $-12x$<br>o $4x$ o $-4x$ o $24x$ o $-24x$ | $-24$ |

   d. Discuss with your partner how using the factors of $-24$, and the value of the coefficient of the $b$-term, $-5$, is helpful when factoring the trinomial $x^2 - 5x - 24$. Summarize your discussion.
   *Answers will vary. I can use the factors to determine which ones, when added together, will result in the middle term.*

*Students may struggle with question 7, since there are no algebra tiles presented already. Consider providing students with the tiles (or allowing them to use them if they already have them) to think through question 7 with their partners.*

*How does the diagram relate to the algebra-tile diagrams used up to this point?*
*Describe if switching where the number of positive and negative x-tiles are placed could give the same answer.*

*Listen as students discuss their thinking in response to part d to determine student thinking to highlight or misconceptions to address when transitioning to 7.2.3 Guided Practice.*

# 7.2.3 Guided Practice: Factoring Trinomials

***Component Overview:*** *Students will practice factoring trinomials where the leading coefficient is one.*

## 7.2.3 Guided Practice: Factoring Trinomials

1. Complete the table for $x^2 + 18x - 40$.

| | $x$ | $-2$ |
|---|---|---|
| $x$ | $x^2$ | ○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$<br>○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$<br>○ $4x$ ○ $-4x$ ○ $20x$ ○ $-20x$<br>○ $5x$ ○ $-5x$ ○ $40x$ ○ $-40x$ |
| $+20$ | ○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$<br>○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$<br>○ $4x$ ○ $-4x$ ○ $20x$ ○ $-20x$<br>○ $5x$ ○ $-5x$ ○ $40x$ ○ $-40x$ | $-40$ |

2. Complete the table for $x^2 + 12x + 32$.

| | $x$ | $+4$ |
|---|---|---|
| $x$ | $x^2$ | ○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$<br>○ $2x$ ○ $-2x$ ○ $16x$ ○ $-16x$<br>○ $4x$ ○ $-4x$ ○ $32x$ ○ $-32x$ |
| $+8$ | ○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$<br>○ $2x$ ○ $-2x$ ○ $16x$ ○ $-16x$<br>○ $4x$ ○ $-4x$ ○ $32x$ ○ $-32x$ | $32$ |

*Use observational data from the 7.2.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 or pulling a small group as they work through the problems to address common misconceptions shown in the 7.2.2 Exploration.*

*For this activity, consider allowing students to complete using a Think-Pair-Share protocol. This will allow students to see what they know on their own and then engage in a collaborative conversation with their partner to clear up potential misconceptions.*

**Component Overview**: *Students will independently practice factoring a trinomial where the leading coefficient is one.*

## 7.2.4 Your Turn!

1. Complete the table for $x^2 - 11x + 18$.

| | $x$ | $-9$ |
|---|---|---|
| $x$ | $x^2$ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$<br>○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$ |
| $-2$ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$<br>○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$ | 18 |

*Monitor students as they work independently through this component. Consider making note of which students may still be struggling, as this informal data can be used when you assign partners in the next lesson.*

*MA.K12.MTR.2.1: Multiple Representations*
*Encourage students to use algebraic phrases, algebra tiles, or other models and diagrams to represent either their process or their thinking.*

# 7.2.5 Wrap-Up: Reflection

***Component Overview**: Students will complete a self-reflection that has them determine how "in the zone" they feel about their understanding of factoring trinomials.*

## 7.2.5 Wrap-Up: Reflection

1. How "in the zone" do you feel right now about factoring trinomials?
   *Answers will vary.*


*If students are unsure how to begin, consider using the learning target or benchmark from the lesson as sentence frames for students to construct their responses.*

# 7.3 Factoring a Trinomial With a Leading Coefficient of 1

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Target | • I can factor a trinomial with a leading coefficient of 1. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

| | |
|---|---|
| Essential Questions | • What methods can you use to factor a trinomial with a leading coefficient of one?<br>• What processes can be used to factor a trinomial? |
| Lesson Narrative | In this lesson, students continue their learning of factoring trinomials with a leading coefficient of one. Students will begin by collaborating on different methods for factoring trinomials, including using the area model. They will have the opportunity to learn how to factor trinomials with a leading coefficient of one using different methods and end the lesson by completing the factoring with the method of their choice. |

| | | | | |
|---|---|---|---|---|
| Component Summary | 7.3.1 Warm-Up (~5 min.) | Factor a trinomial *(SE p. 370)* | 7.3.2 Collaboration (15-20 min.) | Explore different methods of factoring, including the area model and factoring by grouping *(SE p. 370)* |
| | 7.3.3 Guided Instruction (10-15 min.) | Formal instruction on factoring trinomials where $a = 1$ *(SE p. 373)* | 7.3.4 Your Turn! (10 min.) | Independent practice factoring trinomials using preferred method *(SE p. 374)* |
| | 7.3.5 Wrap-Up (~5 min.) | Lesson summary writing activity *(SE p. 374)* | | |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | <u>Key Terms:</u><br>• Area model<br>• Quadratic expression (equation)<br><br><u>Additional Terms:</u> N/A | • N/A | • N/A |

| Teacher Tips | Common Misconceptions | Things to Consider |
|---|---|---|
| | • Students may believe only one method of factoring can be used in specific situations.<br>• Students may attempt to factor without considering the $a$ term in quadratic expressions. | • This lesson is a continuation of student learning in the prior lesson. Students are going to explore the methods of the area model and factoring by grouping. Pay close attention to how students perform in each method to help them determine which method best suits them. |
| | Literacy and Language Routines | Mathematical Thinking and Reasoning (MTR) Standard |
| | **Compare and Connect**<br>7.3.3 Guided Instruction provides students with formal instruction on factoring trinomials where $a = 1$. When teaching students the different methods, consider leveraging this routine to share your thinking while working through the problems in the component. This will support students in making connections between the various representations as they develop their understanding of factoring. | **MA.K12.MTR.2:1 Multiple Representations**<br>7.3.2 Collaboration introduces students to different methods for factoring trinomials with a leading coefficient of one. Students will explore multiple methods such as grouping and area model. In this component, students progress from modeling problems with objects and drawings to using abstract representations. |
| Differentiate Instruction | Since this lesson is an extension of the learning in the prior lesson, consider creating student pairs or groups based on formative data from the prior lesson.<br><br>Utilize physical or digital algebra tiles for students to model and represent their thinking. Students can choose between algebra tiles, area model, grouping, or mental math depending on their unique strengths. Highlight differing approaches to continue facilitating a learning environment that prioritizes *process* over *product* as students develop math thinking and reasoning skills. | |

Lesson Notes:

# 7.3.1 Warm-Up: Bell Work

***Component Overview**: Students will complete a Bell Work activity where they factor a trinomial with a leading coefficient of one.*

### 7.3.1 Warm-Up: Bell Work

1. Complete the table for $x^2 + 3x - 18$.

| | $x$ | $-3$ |
|---|---|---|
| $x$ | $x^2$ | o $1x$ o $-1x$ o $6x$ o $-6x$<br>o $2x$ o $-2x$ o $9x$ o $-9x$<br>o $3x$ o $-3x$ o $18x$ o $-18x$ |
| $+6$ | o $1x$ o $-1x$ o $6x$ o $-6x$<br>o $2x$ o $-2x$ o $9x$ o $-9x$<br>o $3x$ o $-3x$ o $18x$ o $-18x$ | $-18$ |

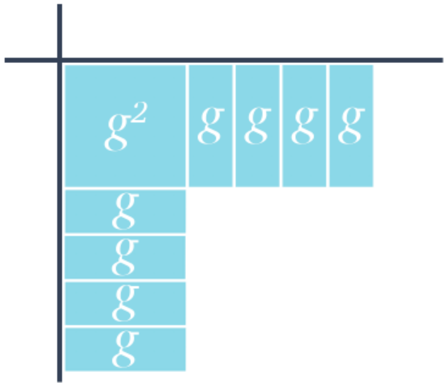
*Think-Pair-Share*
*This task is identical to what students did in the prior lesson. Allow students to have the opportunity to try this one on their own to see what they recall. Then, have students work with a partner to share their answers and come to a consensus. Once partners have had an opportunity to discuss, facilitate a whole-class review of the correct answer if necessary and if time allows.*

# 7.3.2 Collaboration: Different Methods of Factoring

***Component Overview***: *Students will work with a partner or small group to explore different methods of factoring, including area models and grouping.*

## 7.3.2 Collaboration: Different Methods of Factoring

1. Garcelle is trying to factor the trinomial $g^2 + 8g + 12$ using algebra tiles.




Discuss with your partner the mistake that Garcelle made. Write a note to Garcelle to explain her mistake and how she can fix it.

**Means**
**I**
**Start**
**To**
**Acquire**
**Knowledge**
**Experience**
**Skills**

*Answers will vary. Hi Garcelle! Good job recognizing that $4g + 4g$ is $8g$. However, the current setup would require $16$ positive one tiles, which won't match the trinomial. Can you think of factors of $12$ that add up to be $8$? That should help you get back on track!*

*MA.K12.MTR.3.1: Mathematical Fluency*
*Collaboration intentionally highlights three differing methods that produce the same result. Continue highlighting the role of flexibility in fluency, that different representations can provide more pathways that end with the same answer. In the 7.3.3 Guided Instruction, the teacher should continue using visual models if the students are struggling.*

*Question 1 is a recall back to what students did in the prior lesson. Use the data from that lesson to determine which student pairs you may need to visit first to ensure they are on the right track in this component.*

*How many unit tiles will be needed to complete the algebra-tile model Garcelle created? How could you "fix" the g-tiles so that the result will be only 12 unit tiles?*

# 7.3.2 Collaboration: Different Methods of Factoring (continued)

2. Complete the table for $g^2 + 8g + 12$.

| | $g$ | $+2$ |
|---|---|---|
| $g$ | $g^2$ | ○ $1g$ ○ $-1g$ ○ $4g$ ○ $-4g$ ○ $2g$ ○ $-2g$ ○ $6g$ ○ $-6g$ ○ $3g$ ○ $-3g$ ○ $12g$ ○ $-12g$ |
| $+6$ | ○ $1g$ ○ $-1g$ ○ $4g$ ○ $-4g$ ○ $2g$ ○ $-2g$ ○ $6g$ ○ $-6g$ ○ $3g$ ○ $-3g$ ○ $12g$ ○ $-12g$ | 12 |

3. What are the two binomial factors for $g^2 + 8g + 12$?

$g^2 + 8g + 12 = (g + 2)(g + 6)$

Monitor for students who are stuck or struggling in small groups and use inquiry to prompt discussion.

- *How can you determine if the signs should be positive or negative?*
- *What is the relationship between the 12 and the number of g-tiles?*
- *Can the factors listed in the table be reversed to result in the same product?*

Felix showed Garcelle how to use the ***area model*** to factor $g^2 + 8g + 12$.

The ***area model*** is a rectangular diagram that utilizes the decomposition of side lengths by place value to multiply numbers using the distributive property.

4. Fill in the missing information for the ***area model*** for $g^2 + 8g + 12$.

| | $1g$ | 6 |
|---|---|---|
| $1g$ | $g^2$ | $6g$ |
| 2 | $2g$ | 12 |

*Consider creating an anchor chart that represents a trinomial being factored using different methods but arriving at the same answer. Use different colors or other ways to distinguish one method from another easily.*

*The methods included could be using algebra tiles, the area model, or factoring by grouping.*

# 7.3.2 Collaboration: Different Methods of Factoring (continued)

5. Garcelle asked Felix how to determine the monomial that belongs in the orange boxes when creating the area model for $k^2 - 3k - 28$. Felix said that he rewrites $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$ because $-7 \cdot 4 = -28$ and $-7 + 4 = -3$. He knows that one of the orange boxes should result in the product $-7k$ and the other should result in the product $4k$.

   a. Complete the area model for $k^2 - 3k - 28$.

| | $k$ | $-7$ |
|---|---|---|
| $k$ | $k^2$ | $-7k$ |
| $4$ | $4k$ | $-28$ |

   b. What are the two binomial factors for $k^2 - 3k - 28$?

   $k^2 - 3k - 28 = (k + 4)(k - 7)$

   c. Lori uses a different method of factoring. Like Felix, she rewrote $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$, but instead paired the terms as shown.

$$(k^2 - 7k) + (4k - 28)$$

   Work with your partner to finish factoring using Lori's method, known as **factoring by grouping**, using the following steps.

| $(k^2 - 7k) + (4k - 28)$ | |
|---|---|
| Factor out the greatest common factor from each pair of terms. | $k(k - 7) + 4(k - 7)$ |
| The binomial, $k - 7$, is common so the sum can be rewritten. | $(k - 7)(k + 4)$ |

*Consider allowing students to do a whole-class share out to discuss their findings and processes after completing all parts of question 5. Have them compare what they did with the area model in part a with their method of grouping in part c.*

*Consider creating a characteristics T-chart to have students share out features of both methods during their discussion.*

*If Lori grouped the expression as $(k^2 + 4k) + (-7k - 28)$, show if the factors will be the same.*

There are many different methods for factoring trinomials. Relating factoring to algebra tiles or the area model is useful for understanding why the factors of a trinomial of the form $ax^2 + bx + c$ are two binomials.

Using algebra tiles or the area model is also helpful in understanding the relationship the factors of the $c$-term have with the $b$-term.

6. Discuss with your partner the relationship the factors of the $c$-term have with the $b$-term based on the expressions explored in the previous problems.
   *Answers may vary. The $b$ term is the sum of the factors of the $c$ term.*

*MA.K12.MTR.3.1: Mathematical Fluency*
*Different methods for factoring can be helpful for different students. Showing all methods helps to solidify understanding as well as provide insight into which methods make more or less sense to each of your students. Consider ways to leverage their different understandings to provide differentiated support through the remainder of this lesson and unit.*

*Consider adding a debate activity here. Have students split into two groups. In one group, they are for using the area model, and the other group is for using the grouping method. Each team must provide reasoning and rationale on why their method is the best method to use when factoring trinomials.*

# 7.3.3 Guided Instruction: Factoring Trinomials Where $a = 1$

***Component Overview:*** *Students will receive formal instruction on factoring trinomials where a = 1 using different methods.*

## 7.3.3 Guided Instruction: Factoring Trinomials where $a = 1$

A ***quadratic expression***, or ***quadratic equation***, is a polynomial expression or equation that contains a term of degree 2, but no term of higher degree.

Factoring quadratic expressions in the form of trinomials such as $ax^2 + bx + c$, where $a = 1$, results in the factors $(x + g)(x + h)$ where, $g \cdot h = c$ and $g + h = b$. Later in this unit, factoring trinomials where $a$ is a value other than 1 will be explored.

1. Consider the trinomials $y^2 + 14y + 24$, $y^2 - 14y + 24$, $y^2 + 10y - 24$, and $y^2 - 10y - 24$.
   a. Factor each trinomial.

| $y^2 + 14y + 24$ | $y^2 - 14y + 24$ | $y^2 + 10y - 24$ | $y^2 - 10y - 24$ |
|---|---|---|---|
| $(y + 2)(y + 12)$ | $(y - 12)(y - 2)$ | $(y - 2)(y + 12)$ | $(y - 12)(y + 2)$ |

   b. Describe any similarities between the four trinomials.

   *All four trinomials have a constant value of 24 or −24 .*

   c. Describe any similarities between the factored forms of the four trinomials.

   *All four binomial factors have the numbers 12 and 2 as factors.*

*Teachers are encouraged to continue using different methods of factoring throughout this component while continuing to avoid “tricks” or shortcuts. Some students with a strong knowledge of factors will be able to determine the correct pair of factors without scaffolding. Other students will need scaffolding. Avoid insisting on only one method for scaffolding, as this can leave students without a method that supports their thinking.*

*Consider splitting the class based on their preferences for factoring from the prior component. Then they can work in partners to attempt part a before finding a partner from a different group and comparing and contrasting both their process and their product.*

# 7.3.3 Guided Instruction: Factoring Trinomials Where $a = 1$ (continued)

2. Consider the trinomial $x^2 - 2x - 8$.

   a. Factor $x^2 - 2x - 8$ using the area model.

| | $x$ | $-4$ |
|---|---|---|
| $x$ | $x^2$ | $-4x$ |
| $2$ | $2x$ | $-8$ |

   b. Factor $x^2 - 2x - 8$ using factoring by grouping.

   $x^2 - 4x + 2x - 8$
   $(x^2 - 4x) + (2x - 8)$
   $x(x^2 - 4x) + 2(2x - 8)$
   $x(x - 4) + 2(x - 4)$
   $(x - 4)(x + 2)$

   c. Discuss with your partner how to check that the two binomials of the factored form are correct.

   *The distributive property can be applied to the factored binomials to check if the product is the same as the original trinomial.*

*Consider creating a space on the whiteboard, chart paper, or other central location to show how this trinomial was factored using more than one method. Allow different students to come up to the board to do the first step of one method and have them "tag" another student to do the next step. Repeat this again for the other method until it is also factored. Then ask students to compare similarities and differences between the two methods.*

*Tricks and shortcuts lead to misconceptions. Avoid having students memorize phrases like "factors of $c$ whose sum is $b$" and instead use area model and grouping to represent their thinking. While students can use a strategy like finding factors of $(a)c$ whose sum is $b$, this approach should not be a driving instructional focus.*

# 7.3.4 Your Turn!

***Component Overview**: Students will practice factoring quadratic expressions using the method of their choice. For this activity, allow the students to complete independently to practice what they've learned.*

## 7.3.4 Your Turn!

1. Factor each quadratic expression using the method of your choice.

   a. $d^2 + 11d + 24$
   $(d + 3)(d + 8)$

   b. $a^2 - 9a + 14$
   $(a - 7)(a - 2)$

   c. $x^2 + 4x - 96$
   $(x - 8)(x + 12)$

*Consider having students find another student for each problem that factored it using a different method. With this partner, have them compare their answers and come to a consensus if their answers do not match. Once they've gotten their answers, have students initial that they've been used for that problem. Repeat this process for each part of the question to allow students the opportunity to work with different students throughout the review of this component.*

# 7.3.5 Wrap-Up: Cool Down

***Component Overview**: Students will write a note to someone who missed class that summarizes the lesson. This activity is designed to be completed independently to allow students the opportunity to respond to an open-ended question about what they learned.*

### 7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson. Include something you learned and a mistake you learned from.
   *Answers may vary.*

*Make space for creativity and choice in student responses to increase engagement and agency.*

- *Students can use social media to record a short video describing their findings.*
- *Students can tutor a friend or family member and submit a recording of their session or evidence of the work of their friend or family member.*

# 7.4 Factoring a Trinomial With a Leading Coefficient Greater Than 1

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Targets | • I can use algebra tiles to factor a trinomial with a leading coefficient greater than $1$.<br>• I can factor a trinomial with a leading coefficient greater than $1$. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real or complex number system. |

| | |
|---|---|
| Essential Question | • What are the similarities and differences between the process for factoring trinomials with a leading coefficient of one and a leading coefficient greater than one? |
| Lesson Narrative | In this lesson, students continue their work with factoring but will now explore factoring trinomials where the leading coefficient is not one. Students begin by connecting this process to the algebra tiles that they used in earlier lessons before moving to more abstract representations of the process. They will get the opportunity to continue practicing this process throughout the lesson. |

| | | | | |
|---|---|---|---|---|
| Component Summary | 7.4.1 Warm-Up (~5 min.) | Growth mindset activity *(SE p. 375)* | 7.4.4 Try It (10 min.) | Practice factoring trinomials using preferred method *(SE p. 380)* |
| | 7.4.2 Guided Instruction (15 min.) | Instruction on factoring trinomials when a is not equal to one *(SE p. 376)* | 7.4.5 Your Turn! (5-10 min.) | Independent practice on factoring trinomials *(SE p. 381)* |
| | 7.4.3 Guided Practice (10 min.) | Structured practice on factoring trinomials *(SE p. 379)* | 7.4.6 Wrap-Up (~5 min.) | Muddiest Point self-reflection activity *(SE p. 381)* |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> N/A | • N/A | • N/A |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

- Student entry points and ways of thinking may not reflect the representation of area model of factoring as presented in this lesson. For example, allow students to create their own tables in question 6d that best reflect their own way of thinking and representing their process.

</td><td>

- Avoid using “tricks” during this lesson. Students should see consistency through the process of factoring trinomials. They shouldn’t see factoring trinomials when $a = 1$ and when $a \neq 1$ as two different processes. Rather, use the methods previously developed in Lessons 1-3 to extend the trinomials where $a \neq 1$.
- If time is an issue, consider assigning 7.4.5 Your Turn! for homework.

</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>

**Notice and Wonder**

7.4.2 Guided Instruction provides students with guidance on factoring trinomials where $a \neq 1$. Students will review the methods and concepts they’ve learned in prior lessons and apply them to these trinomials. As such, students will be asked to describe what they notice and differences they see as they connect what they already know to this new learning.

</td><td>

**MA.K12.MTR.2.1: Multiple Representations**

This lesson provides opportunities for students to factor trinomials in multiple ways using multiple methods and approaches, such as the area model and grouping. Guide students from concrete to pictorial to abstract representations as understanding progresses. Students will continue to build understanding through modeling and using manipulatives.

</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">

Notice algebra tiles as the focus of one of the lesson’s learning targets. Algebra tiles are one strategy for accomplishing the benchmark focus of rewriting polynomials, but algebra tiles are never a requirement for students to demonstrate. Continue highlighting different strategies throughout this lesson, with algebra tiles acting as one of many approaches for making sense of polynomials and factoring.

- Guided Instruction Focus: Algebra tiles
- Guided Practice Focus: Area model
- Try It: Student choice
- Your Turn!: Area model and factor by grouping

</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.4.1 Warm-Up: Growth Mindset

***Component Overview**: Students will complete a growth mindset activity by reviewing statements from celebrities and responding to growth mindset questions.*

## 7.4.1 Warm-Up: Growth Mindset

1. Each statement describes the celebrity indicated.

| Statement | Celebrity |
|---|---|
| He was never offered a top Division 1 scholarship. Scouting Reports said his explosiveness and athleticism were below standard. | Steph Curry |
| His first book was rejected by 27 different publishers. They said his material was too different and would never sell. | Dr. Seuss |
| He was fired from a newspaper for not being creative enough. They said he lacked imagination and had no good ideas. | Walt Disney |

a. How did a growth mindset help these celebrities?
*Answers will vary. They turned rejection into a drive to work even harder to be successful.*

b. What can you do to maintain a growth mindset if you get frustrated with today's (or any day's) lesson to help you keep a growth mindset?
*Answers will vary. I can use the frustration as fuel to be work harder to be successful with the content.*

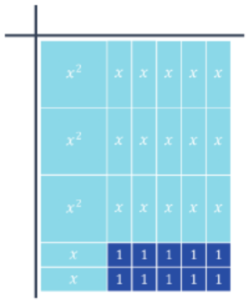
*Make this more about their thinking and their discussion to these questions rather than physically writing their responses in the book. By leveraging this activity as a discussion, you are more likely to get more detailed responses from students.*

# 7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

***Component Overview**: Students will learn how to factor trinomials where the leading coefficient is not one. They will connect their prior learning with the use of algebra tiles, the distributive property, area models, and grouping to learn how to factor this type of trinomial.*

## 7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

1. Use the algebra tiles to determine the two binomials that are factors of $3x^2 + 17x + 10$.




$(3x + 2)(x + 5)$

2. Discuss with your partner any differences you notice between the binomial factors in question 1 and factoring trinomials where $a = 1$. Summarize your discussion.
*Answers will vary. We noticed that there was never more than one $x^2$ tile when $a = 1$.*

3. The steps for applying the distributive property to multiply $(3x + 2)(x - 5)$ are shown.

| Step | $(3x + 2)(x - 5)$ | Result |
|---|---|---|
| Distribute $3x$ to $x$ | $(3x + 2)(x - 5)$ | $3x^2$ |
| Distribute $3x$ to $-5$ | $(3x + 2)(x - 5)$ | $-15x$ |
| Distribute $2$ to $x$ | $(3x + 2)(x - 5)$ | $2x$ |
| Distribute $2$ to $-5$ | $(3x + 2)(x - 5)$ | $-10$ |

*Guide students through this component, making explicit connections to what they have already learned when factoring trinomials where $a = 1$. Consider using Think Aloud and mark-up strategies as you model your thinking.*

*Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or potential misconceptions to address.*

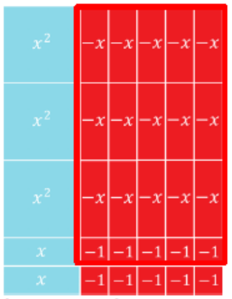
*What is the relationship between the distributive property and factoring?*

*What do you notice is different about the resulting trinomial in part a from the partial products shown in the table?*

# 7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$ (continued)

a. Based on the results in the table, what is the product of $(3x + 2)(x - 5)$?
$3x^2 - 13x - 10$

b. The algebra tiles diagram models the result of $(3x + 2)(x - 5)$.




Draw a rectangle around $-15x$ in the algebra tiles diagram for $3x^2 - 13x - 10$.

*For question 3b, students may have a hard time connecting to the visual diagram in the book while you are providing instruction. Consider using physical tiles or a visual display of your diagram to provide a reference point while you are describing your thinking and steps.*

c. Discuss with your partner how the algebra tiles diagram demonstrates the distribution of $3x$ to $-5$. Summarize your discussion.
*When $3x$ is distributed to $-5$, the result is $-15x$. The $-15$ $x$ tiles represent $-15x$.*

*Give students one to two minutes to discuss question 3c. Allow them the opportunity to share their thinking before moving on to question 4 in this activity. Use question 4 as an "in the moment" prompting question to elicit student responses as a whole group.*

4. Compare the algebra tiles diagram for $3x^2 + 17x + 10$ to the algebra tiles diagram for $3x^2 - 13x - 10$. Describe the differences.
*Answers will vary. One had negative $x$ tiles and one had positive.*

Factoring trinomials of the form $ax^2 + bx + c$, where $a = 1$, resulted in the factored form $(x + g)(x + h)$, where $g \cdot h = c$ and $g + h = b$.

*Prompt students to read and discuss this paragraph. Use mark-up strategies to highlight the connections between the form of the trinomial and its factors to lay foundations for factoring quadratic expressions in Units 8 and 9.*

5. Use the examples in the table to explore factoring trinomials of the form $ax^2 + bx + c$ where $a \neq 1$. Summarize the similarities and differences of the factored forms where $a = 1$ and where $a \neq 1$.

| Trinomial | $3x^2 + 17x + 10$ | $3x^2 - 13x - 10$ |
|---|---|---|
| Factored form | $(3x + 2)(x + 5)$ | $(3x + 2)(x - 5)$ |
| Similarities | *Answers will vary.* | |
| Differences | *Answers will vary.* | |

When factoring quadratics of the form $ax^2 + bx + c$, the coefficient $b$ is the sum of factors of the product of $a$ and $c$.

6. In $3x^2 + 17x + 10$ and $3x^2 - 13x - 10$, the products of $a$ and $c$ are $30$ and $-30$, respectively.

   a. List all the pairs of factors for both $30$ and $-30$.

| Value | Pairs of Factors |
|---|---|
| $30$ | $1, 30; 2, 15; 3, 10; 5, 6$ |
| $-30$ | $-1, 30; -2, 15; -3, 10; -5, 6$<br>$1, -30; 2, -15; 3, -10; 5, -6$ |

   b. Circle the factor pair for $30$ whose sum is $17$.
   $2, 15$

   c. Circle the factor pair for $-30$ that whose sum is $-13$.
   $2, -15$

*What is the impact of having a negative value as part of this process?*

*Why is there double the amount of factor pairs for $-30$ than there is for $30$? How can that play a role in the factoring process?*

# 7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$ (continued)

d. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial expression.

| $3x^2 + 17x + 10$ | $3x^2 - 13x - 10$ |
|---|---|
| $3x^2 + 2x + 15x + 10$ | $3x^2 + 2x - 15x - 10$ |
| $3x^2 + 15x + 2x + 10$ | $3x^2 - 15x + 2x - 10$ |
| $3x(x + 5) + 2(x + 5)$ | $3x(x - 5) + 2(x - 5)$ |
| $(3x + 2)(x + 5)$ | $(3x + 2)(x - 5)$ |

*Filling in the table in part d may not fit within the working schema of all students. The first line is an expansion of the expression using the factor pair. The second line is the same expansion but with the terms in a different order. This is to highlight the use of the commutative property, as students can factor using either order. In this example, the preselected work used them with the $15x$ and the $2x$. It is important for students to know that they would arrive at the same answer if they used Line 1 to factor by grouping.*

# 7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$

***Component Overview***: *Students will engage in guided practice with factoring trinomials with a leading coefficient greater than one through both grouping and the use of the area model.*

## 7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$

1. Factor $5x^2 + 6x - 8$ by grouping.
   $(5x - 4)(x + 2)$

2. Consider the trinomial $3x^2 - 13x - 10$.
   a. Complete the table by choosing the appropriate values.

| | | |
|---|---|---|
| | $3x$ | ○ $1$ ○ $-1$ <br> ○ $2$ ○ $-2$ <br> ○ $5$ ○ $-5$ <br> ○ $10$ ○ $-10$ |
| $x$ | $3x^2$ | ○ $1x$ ○ $-1x$ <br> ○ $2x$ ○ $-2x$ <br> ○ $5x$ ○ $-5x$ <br> ○ $10x$ ○ $-10x$ |
| ○ $1$ ○ $-1$ <br> ○ $2$ ○ $-2$ <br> ○ $5$ ○ $-5$ <br> ○ $10$ ○ $-10$ | ○ $3x$ ○ $-3x$ <br> ○ $6x$ ○ $-6x$ <br> ○ $15x$ ○ $-15x$ <br> ○ $30x$ ○ $-30x$ | $-10$ |

   b. Use the values to complete the area model.

| | | |
|---|---|---|
| | $3x$ | $2$ |
| $x$ | $3x^2$ | $2x$ |
| $-5$ | $-15x$ | $-10$ |

*Allow informal observation throughout the 7.4.2 Guided Instruction to determine how "guided" this Guided Practice should be for your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group. There are additional practice components to follow, so providing this additional support can solidify understanding prior to releasing students in the next component.*

*Students may get confused on the role of the values in the table. Consider explaining how the monomials in the middle row on the right are the product of $x$ and the values in the column header and how the monomials in the bottom row in the middle are the product of $3x$ and the values in the row header.*

*Which method did you find easier to use for this trinomial? Does that preferred method match what you previously preferred to use when factoring?*
*How can you ensure that the values maintain their correct signs?*

# 7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$ (continued)

3. Factor $5x^2 - 3x - 8$ using the area model.

| | $x$ | $1$ |
|---|---|---|
| $5x$ | $5x^2$ | $5x$ |
| $-8$ | $-8x$ | $-8$ |

$(5x - 8)(x + 1)$

*Consider the use of a Think-Pair-Share protocol for question 3. Students should have a couple of minutes to try it on their own and then work with a partner to compare answers and discuss their processes in order to come to a consensus on the final answer.*

# 7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

***Component Overview***: *Students will practice factoring trinomials where the leading coefficient is not one. For this activity, students can work independently or in pairs.*

### 7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

1. Factor $2z^2 - 21z - 36$.
   $(2z + 3)(z - 12)$

2. Factor $3b^2 - 23b + 14$.
   $(3b - 2)(b - 7)$

*For this activity, there is no specific direction provided on the way to factor these expressions. Encourage students to use the method of their choice and find a partner who chose a different method to compare processes to make meaningful connections between representation and concept.*

*Consider having students factor using more than one method to add additional practice and build fluency.*

# 7.4.5 Your Turn!

***Component Overview****: Students will practice factoring a quadratic expression using both the area model and by grouping.*

## 7.4.5 Your Turn!

1. Consider the quadratic expression.

$$5x^2 + 21x + 18$$

a. Factor using the area model.

| | $5x$ | $6$ |
|---|---|---|
| $x$ | $5x^2$ | $6x$ |
| $3$ | $15x$ | $18$ |

b. Factor by grouping.

$5x^2 + 6x + 15x + 18$
$(5x^2 + 6x) + (15x + 18)$
$x(5x + 6) + 3(5x + 6)$
$(x + 3)(5x + 6)$

*Monitor students as they work independently in this component to determine where students may need additional differentiated support.*

*Encourage students to use the method of their choice if they struggle with area model or factoring by grouping. If they choose a different method like algebra tiles or guess and check, pair them with a student who used area model to compare and make connections to begin building conceptual connections between methods.*

# 7.4.6 Wrap-Up: Reflection

***Component Overview***: *Students will reflect on what they found to be the most difficult or confusing point in this lesson.*

## 7.4.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on factoring trinomials where $a \neq 1$?
   *Answers will vary.*

*What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding related to factoring and how to best address potential gaps in understanding throughout the remainder of the unit.*

# 7.5 Factoring Trinomials

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Target | • I can factor trinomials. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real or complex number system. |

| | |
|---|---|
| Essential Question | • How can you apply what you've learned to factor trinomials? |
| Lesson Narrative | In this lesson, students continue their learning with factoring trinomials. Students will engage in a collaborative Exploration relating the terms within given expressions to the factors of the expressions. They will then engage in a culminating card sort task where they will match a quadratic expression with its representations using algebra tiles, area model, expanded form, and factored form. Additionally, students will learn to factor trinomials when the leading coefficient has multiple factors before wrapping up the lesson. |

| | | | | |
|---|---|---|---|---|
| Component Summary | 7.5.1 Warm-Up (~5 min.) | Problem-solving activity about walking the state of Florida *(SE p. 382)* | 7.5.3 Activity (15-20 min.) | Card sort activity matching quadratic expression, algebra tiles, area model, expanded form, and factored form *(SE p. 385)* |
| | 7.5.2 Exploration (10 min.) | Collaborative exploration relating terms in trinomials to factors of the expression *(SE p. 383)* | 7.5.4 Guided Instruction (10 min.) | Formal instruction on factoring trinomials when the leading coefficient has multiple factors *(SE p. 385)* |
| | 7.5.5 Wrap-Up (~5 min.) | Reflect on the lesson using a KWL chart *(SE p. 386)* | | |

| | Glossary | Materials | Additional Materials |
|---|---|---|---|
| Resources | Key Terms: N/A<br><br>Additional Terms: N/A | • Card sort cards<br>• (Optional) Algebra tiles | • For pacing purposes, consider completing the card sort activity on your own to determine the amount of time you want to provide for this activity. |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may attempt to memorize a list of steps to factor, solidifying misconceptions that may include applying methods of factoring that work when $a = 1$ to expressions in which $a \neq 1$.</li></ul></td><td><ul><li>Students are applying what they've learned in the prior lessons in a culminating experience in the 7.5.3 Activity. Consider the best way to group students for this activity based on their performances in the prior lessons, including whether there is a group of students that would benefit from this activity being more guided.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns<br>7.5.3 Activity has students working on a card sort where they match the quadratic expression to its representation with algebra tiles, area model, expanded form, and factored form. Students can use this instructional routine to take turns in the work of this activity by spotting matches and justifying their answers to their group.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning<br>7.5.3 Activity requires students work with partners or small groups to complete a card sort. Students will need to analyze the problem in a way that makes sense given the task. They will also need to stay engaged and maintain a positive mindset when working to solve the task in this activity. Consider ways to recognize students' efforts when solving challenging problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 7.5.2 Exploration, students will work with a visual diagram that represents the use of algebra tiles. Consider providing physical algebra tiles or paper cut-out representations to allow for a hands-on Exploration.<br><br>7.5.3 Activity is a collaborative card sort that can be done with partners or small groups. Consider grouping students based on their performances in the prior factoring lessons. If data is available, consider creating groups based on their strengths in each method for a dynamic learning experience for all the students in that group.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.5.1 Warm-Up: Walking Florida

***Component Overview***: *Students will be presented with some data about the state of Florida. They will use this data to answer questions to apply mathematical knowledge to solve a problem.*

## 7.5.1 Warm-Up: Walking Florida

*Consider employing a Think-Pair-Share routine for this activity. Allow students a few minutes to try this on their own. Then have them work with a partner to share their work before coming to a consensus on the correct answer prior to reviewing the answer as a whole group.*

1. Florida has the longest coastline, 1197 miles, in the contiguous United States. The distance from Key West to the farthest west Florida border on I-10 is 839 miles. The average person walks 3 miles per hour (mph).

   a. How many hours would it take someone walking 3 mph to walk from Key West to Florida's western border?

   $839 \div 3 = 279.\overline{666}$

   *It would take about 280 hours.*

   b. How much longer, in days, would it take someone to walk along Florida's coastline than to walk from Key West to Florida's western border?

   $(1197mi - 839mi) = 358mi$

   $358\ mi \cdot \frac{1\ hour}{3\ mi} \cdot \frac{1\ day}{24\ hours} = 4.97\bar{2}\ days$

   *It would take approximately 5 more days for someone to walk along the entire coastline than for them to just walk from Key West to Florida's western border.*

# 7.5.2 Exploration: Relating Terms and Factors

***Component Overview***: *Students will engage in a collaborative Exploration relating terms and factors. In this activity, students will determine the quadratic expression that results after the multiplication of two binomials. Students will then practice using algebra tiles to factor trinomials. This activity is designed to be completed with a partner or small group.*

## 7.5.2 Exploration: Relating Terms and Factors

1. The steps for using the distributive property to multiply $(2x + 3)(2x - 2)$ are shown.

| Step | $(2x + 3)(2x - 2)$ | Result |
|---|---|---|
| Distribute $2x$ to $2x$ | $(2x + 3)(2x - 2)$ | $4x^2$ |
| Distribute $2x$ to $-2$ | $(2x + 3)(2x - 2)$ | $-4x$ |
| Distribute $3$ to $2x$ | $(2x + 3)(2x - 2)$ | $6x$ |
| Distribute $3$ to $-2$ | $(2x + 3)(2x - 2)$ | $-6$ |

a. Based on the results in the table, what quadratic expression is the product of $(2x + 3)(2x - 2)$?
$4x^2 + 2x - 6$

b. Tyra and Min-Joon were discussing how to find the $x$-term of the quadratic expression for $(2x + 3)(2x - 2)$.

| Tyra | Min-Joon |
|---|---|
| The $x$-term of $(2x + 3)(2x - 2)$ can be quickly found by adding $3$ and $-2$. | Looking at the table, the $x$-term of the quadratic expression is $2x$. I think you could just use what repeats. |

Discuss with your partner both students' strategies.
*Answers will vary.*

c. Both students have made a mistake. Describe to Tyra and Min-Joon how to find the $x$-term of the quadratic expression for $(2x + 3)(2x - 2)$.
*Distribute each term in the first binomial with each term in the second. Then, combine like terms.*

*Use student responses from Lesson 4 Wrap-Up or this lesson's Warm-Up. Strategically group students for both this activity and the Card Sort.*

*What do you notice about the methods Tyra and Min-Joon used and what was found in the table you just completed?*

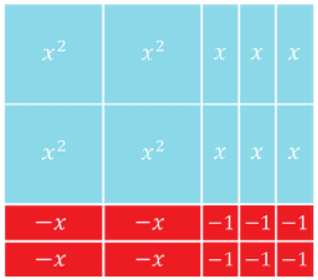
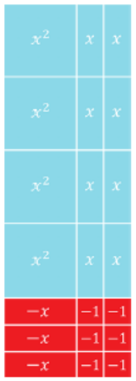
*Consider having students actually try the original methods that Tyra and Min-Joon suggested to see what answers would have been produced. Use this as a springboard for conversation around the errors and solutions identified in part c.*

# 7.5.2 Exploration: Relating Terms and Factors (continued)

d. Discuss with your partner whether or not the $x$-term of the quadratic expression $(4x-3)(x-2)$ is different from the $x$-term of the quadratic expression $(2x-3)(2x-2)$. Summarize your discussion.
*No, the first trinomial is $4x^2-11x+6$, whereas the second is $4x^2-10x+6$, so although the first and third terms are the same, the middle terms are not.*

2. Consider the trinomials $4x^2+2x-6$ and $4x^2+5x-6$.

a. Use the algebra tiles to factor each trinomial.

| **Trinomial** | $4x^2+2x-6$ | $4x^2+5x-6$ |
|---|---|---|
| **Algebra Tiles** |  |  |
| **Factored Form** | $(2x-2)(2x+3)$ *or* $2(x-1)(2x+3)$ | $(4x-3)(x+2)$ |

b. The trinomials have the same leading term, $4x^2$. List the two ways $4x^2$ is factored.
$x$, $4x$ *and* $2x, 2x$

c. List the factor pairs of $-24$.
$1,-24; 2,-12; 3,-8; 4,-6; -1,24; -2,12; -3,8; -4,6$

d. Which pair of factors of $-24$ add to $2$?
$-4,6$

*What terms are being multiplied to find the x-term? How can knowing that help you quickly determine if the x-terms will be different?*

*If available, consider providing students with physical algebra tiles to use when answering question 2. If they are not available, consider creating paper cut-out versions of what's needed for these trinomials to allow for a hands-on experience.*

*What do you notice is different about the first trinomial's factored form?*

*What are some of the challenges when working with factor pairs of a negative number?*

# 7.5.2 Exploration: Relating Terms and Factors (continued)

e. Discuss with your partner how the pair of factors of $-24$ that add to $2$ is related to the factored form of $4x^2 + 2x - 6$. Show the factors along the algebra tiles.

*Answers will vary.* $2x$ *is the sum of the products of the innermost and outermost terms in the binomial factors.*

| | $x$ | $x$ | $1$ | $1$ | $1$ |
|---|---|---|---|---|---|
| $x$ | $x^2$ | $x^2$ | $x$ | $x$ | $x$ |
| $x$ | $x^2$ | $x^2$ | $x$ | $x$ | $x$ |
| $-1$ | $-x$ | $-x$ | $-1$ | $-1$ | $-1$ |
| $-1$ | $-x$ | $-x$ | $-1$ | $-1$ | $-1$ |

*Create a space for whole-group discussion of part e as it relates to what they've done in question 2. The knowledge from this Exploration will assist them in their work in the upcoming card sort activity.*

# 7.5.3 Activity: Card Sort

***Component Overview:*** *Students will work in partner pairs or small groups to complete a card sort activity. In this activity, students will match the card that has a quadratic expression on it with its corresponding representation using algebra tiles, area model, expanded form, and factored form.*

## 7.5.3 Activity: Card Sort

1. Work with your partner to sort the cards. Write the letter of the matching card for each representation.

| Quadratic Expression | Algebra Tiles | Area Model | Expanded Form | Factored Form |
|---|---|---|---|---|
| $6x^2 - x - 2$ | P | D | T | F |
| $6x^2 + x - 2$ | K | O | Z | B |
| $6x^2 - 7x + 2$ | E | U | Y | G |
| $6x^2 - 11x - 2$ | R | J | A | M |
| $6x^2 + 11x - 2$ | V | S | L | H |
| $6x^2 - 11x + 2$ | N | I | W | C |

2. Describe one strategy you and your partner used to sort the cards.
   *Answers will vary. My partner and I started with the tiles first because we are more comfortable with that. My partner and I started with the expanded form because that was the easiest card to eliminate from the stack.*

*Pre-cut and group cards into plastic bags with a group number on the outside to pass out to each group. To reduce loss of instructional time, consider having students work with the same partner or small group that they worked with in the previous component.*

*How students respond to each matching category in this activity can provide insight into misconceptions and areas that need additional support prior to the end of the unit.*

*Consider creating an index card or printout with the correct letters for each row. Keep this with you as you circulate the room to better provide just-in-time feedback to students.*

*Teachers may have algebra tiles on hand to help students manipulate tiles or use a virtual set of algebra tiles. The teacher may wish to modify the activity if time is a concern. The best modification may be to give students a few of the correct matches. Other modifications might take away from the integrity of students making their own connections.*

# 7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when $a$ Has Multiple Factors

***Component Overview***: *Students will receive guided instruction on how to factor trinomials when the leading coefficient has multiple factors.*

## 7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when $a$ has Multiple Factors

Factoring trinomials of the form $ax^2 + bx + c$, where the value of $a$ has multiple factors, requires all of the factors of $a$ to be considered when factoring using algebra tiles or the area model.

1. Complete the table for $10x^2 + x - 3$.

| | $2x$ | $-1$ |
|---|---|---|
| $5x$ | $10x^2$ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$<br>○ $3x$ ○ $-3x$ ○ $15x$ ○ $-15x$<br>○ $5x$ ○ $-5x$ ○ $30x$ ○ $-30x$ |
| $3$ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$<br>○ $3x$ ○ $-3x$ ○ $15x$ ○ $-15x$<br>○ $5x$ ○ $-5x$ ○ $30x$ ○ $-30x$ | $-3$ |

2. Which expanded form of $10x^2 + x - 3$ could be used to factor the expression by grouping?

   - ○ $10x^2 + 5x - 6x - 3$
   - ○ $10x^2 - 5x + 6x - 3$
   - ○ $10x^2 - 10x + 3x - 3$
   - ○ $10x^2 + 10x - 3x - 3$

3. Use the expanded form to factor $10x^2 + x - 3$.
   $(2x - 1)(5x + 3)$

*Consider the placement of this activity in the lesson. Students will have just finished a collaborative and potentially lively discussion during the card sort activity.*

*To maintain instructional momentum, consider allowing students to remain with their partner or small group and use this material in a collaborative release structure. For example, read the narrative prior to question 1 and release students to try question 1 with their partner or group and then review as a whole group.*

*Repeat this process for questions 2 and 3 or allow students to independently practice on their own while pulling a small group for accelerated learning.*

# 7.5.5 Wrap-Up: Reflection

***Component Overview***: *Students will complete a reflection activity regarding the 7.5.2 Exploration activity. This activity is designed to be done independently to collect qualitative student data on their experiences with the 7.5.2 Exploration activity.*

## 7.5.5 Wrap-Up: Reflection

1. Complete the table.
   *Answers will vary.*

| The terms and information used in the exploration and activity that I knew were … | Something I wondered during the exploration or activity was … | Something I learned during the exploration or activity was … |
|---|---|---|
| | | |

*Consider pairing students to discuss their reflections to see if peers can answer any of their questions or wonderings.*

# 7.6 Factoring by Grouping

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Target | • I can factor a polynomial by grouping. |

| Benchmark Coherence | Building On | Working Toward |
|---|---|---|
| | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

| | |
|---|---|
| Essential Question | • How can you determine if factoring by grouping is the best method for factoring a polynomial? |
| Lesson Narrative | In this lesson, students explore the process of factoring by grouping. Students will learn how this method can be used when the coefficient of a term has multiple factors and when a polynomial has more than three terms. They will have opportunities throughout the lesson to practice factoring by grouping with a partner and independently, including working collaboratively to consider the process of factoring given polynomials. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 7.6.1 Warm-Up (~5 min.) | Notice and Wonder activity with Coral Castle *(SE p. 387)* | 7.6.4 Try It (10 min.) | Collaborative practice with factoring by grouping *(SE p. 389)* |
| | 7.6.2 Guided Instruction (10-15 min.) | Formal instruction on factoring by grouping *(SE p. 387)* | 7.6.5 Your Turn! (5 min.) | Independent practice with factoring by grouping *(SE p. 390)* |
| | 7.6.3 Guided Practice (10 min.) | Guided practice with factoring by grouping *(SE p. 389)* | 7.6.6 Wrap-Up (5-10 min.) | Ticket Out the Door with factoring *(SE p. 390)* |

| Resources | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| | Key Terms: N/A<br><br>Additional Terms: N/A | • N/A | • N/A |

| | Common Misconceptions | Things to Consider |
|---|---|---|
| Teacher Tips | • Students may not recognize when a trinomial can't be factored and is prime.<br>• Students may try to factor by grouping when it is not possible. | • If time is an issue, consider assigning 7.6.5 Your Turn! for homework to allow adequate time for students to complete the 7.6.6 Wrap-Up for use in planning differentiated support before the end of the unit. |
| | **Literacy and Language Routines** | **Mathematical Thinking and Reasoning (MTR) Standard** |
| | **Notice and Wonder**<br>7.6.1 Warm-Up provides students with an opportunity to determine what they notice and what they wonder about the sculptures at Coral Castle in Miami, Florida. | **MA/K12.MTR.3.1: Mathematical Fluency**<br>7.6.3 Guided Practice and 7.6.5 Your Turn! have students factoring polynomials by grouping. During these, students will select efficient and appropriate methods for solving problems. Additionally, they will complete these tasks accurately and with confidence. The lesson offers multiple opportunities for students to practice efficient and generalizable methods. |
| Differentiate Instruction | Consider creating an anchor chart for student reference that highlights the process explored in 7.6.4 Try It with the provided example and then again with the example from question 2 from activity 7.6.3 Guided Practice. | |
| Lesson Notes: | | |

# 7.6.1 Warm-Up: Notice and Wonder

***Component Overview:*** *Students will review an image from Coral Castle in Miami, FL, to answer some questions about what they notice and wonder.*

## 7.6.1 Warm-Up: Notice and Wonder

1. Coral Castle in Miami, Florida, was created by Edward Leedskalnin. He carved over 1100 tons of coral rock using an unknown process, often carving at night.

   a. What do you notice about the sculptures?
   *Answers will vary. I notice that there are multiple different shapes. I notice that some are surrounded by fencing and some are not.*

   b. What do you wonder about the process Ed used?
   *Answers will vary. I wonder how he decided what shapes he would sculpt. I wonder how he knew what he could make out of that amount of coral rock.*

*After students have had an opportunity to view the image and answer the questions on their own, ask several students to share the things they noticed and things they wondered.*

# 7.6.2 Guided Instruction: Factoring by Grouping

***Component Overview:*** *Students will delve deeper into factoring by grouping. Instruction includes factoring polynomials with more than three terms using this method.*

## 7.6.2 Guided Instruction: Factoring by Grouping

When factoring $ax^2 + bx + c$, the coefficient $b$ is the sum of factors of the product of $a$ and $c$.

1. In both $8x^2 - 19x - 15$ and $8x^2 - 2x - 15$, the product of $a$ and $c$ is $-120$.

   a. List all the pairs of factors for $-120$.
   $-1,120; -2,60; -3,40; -4,30; -5,24; -6,20; -8,15; -10,12;$
   $1,-120; 2,-60; 3,-40; 4,-30; 5,-24; 6,-20; 8,-15; \boxed{10,-12;}$

   b. Circle the factor pair for $-120$ that adds to the sum of $-19$.
   $5, -24$

   c. Draw a rectangle around the factor pair for $-120$ that adds to the sum of $-2$.
   $10, -12$

2. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial.

| $8x^2 - 19x - 15$ | $8x^2 - 2x - 15$ |
|---|---|
| $8x^2 - 24x + 5x - 15$ | $8x^2 - 12x + 10x - 15$ |
| | |
| $8x(x - 3) + 5(x - 3)$ | $4x(2x - 3) + 5(2x - 3)$ |
| $(8x + 5)(x - 3)$ | $(4x + 5)(2x - 3)$ |

Factoring by grouping is helpful when factoring trinomials of the form $ax^2 + bx + c$, where the value of $a$ has multiple factors.

*Students will have opportunities to practice in the remaining components of this lesson. The purpose of this component is to provide explicit instruction in factoring by grouping to ensure understanding. Guide students through the questions in this component as they follow along. Consider thinking aloud through each process while using talk moves like "Revoice" to have students make sense of the information in their own words as they formulate working definitions of factoring by grouping.*

*Consider turning question 1 into an interactive partner activity. Have students complete each part with a partner and then chart up all the pairs of factors for $-120$ onto a central location for all students to see and use in question 2.*

*The first two lines of the table can be confusing. The purpose of the first line is to rewrite the expression using the factor pair. The second line highlights that the commutative property can be used to rewrite the expression.*

# 7.6.2 Guided Instruction: Factoring by Grouping (continued)

3. Factor $6x^2 - 19x + 15$ by grouping.
   $6x^2 - 10x - 9x + 15$
   $2x(3x - 5) - 3(3x - 5)$
   $(3x - 5)(2x - 3)$

Factoring by grouping can also be helpful when factoring a polynomial with four terms.

4. Factor $3y^4 - 6y^3 + 9y^2 - 18y$ by grouping.
   $3y(y - 2)(y^2 + 3)$

5. The polynomial $3y^4 + 9y^2 - 6y^3 - 18y$ is a rearranged form of $3y^4 - 6y^3 + 9y^2 - 18y$.

   Factor $3y^4 + 9y^2 - 6y^3 - 18y$ by grouping.
   $3y(y - 2)(y^2 + 3)$

6. What do you notice about the factored forms of $3y^4 - 6y^3 + 9y^2 - 18y$ and $3y^4 + 9y^2 - 6y^3 - 18y$?
   *They are the same.*

*Consider posing one or more of the following discussion questions before wrapping up this component.*

- *How can factoring by grouping be useful when the leading coefficient has multiple factors?*
- *How can factoring by grouping be useful when a polynomial has four terms?*
- *Does writing the polynomials in standard form make a difference when factoring by grouping?*

# 7.6.3 Guided Practice: Factoring by Grouping

***Component Overview**: Students will practice factoring by grouping in a structured environment.*

## 7.6.3 Guided Practice: Factoring by Grouping

1. Factor $6x^2 + x - 35$ by grouping.

   $(3x - 7)(2x + 5)$

2. Factor $-15g + 35gh - 7h + 3$ by grouping.
   $(7h - 3)(5g - 1)$

*Use informal observation from the previous component and lessons to inform how "guided" this Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.*

*When reviewing the answers as a whole group, consider prompting students to elaborate on their process and steps for each question. Position students to answer questions that ask more about the procedures and concepts rather than just "What did you get for your answer?"*

# 7.6.4 Try It: Factoring by Grouping

***Component Overview**: Students will work with a partner to answer questions about the process of factoring by grouping. For this activity, students should work with a partner.*

## 7.6.4 Collaboration: Factoring by Grouping

Work with your partner to complete the following.

1. Factor $2x^2 + 3x + 4x + 12$ by grouping.
   *Not factorable. Prime.*

2. What happens when you try to group the expression and find a common factor?
   *The factored binomials are not equivalent.*

3. If you rearrange the middle two terms, what happens when you try to factor by grouping again?
   *The same results as before.*

4. Complete the statement.

   When factoring an expression by grouping, group terms

   that have coefficients that
   - o are exactly the same.
   - o have common factors.
   - o are in numerical order.

5. Work with your partner to create a strategy for the best way to group an expression.
   *Answers will vary*.

*Release students to work on this activity from start to finish on their own with a partner. Once completed, review each question as a whole class. Engage students in a rich discussion around questions 2-4 to help them better understand the process. Have students then share out the strategies they developed in question 5.*

*For a challenge, have students describe what would happen with this process if the conditions of the incorrect choices in question 4 existed.*

# 7.6.5 Your Turn!

***Component Overview***: *Students will work independently to practice factoring by grouping.*

## 7.6.5 Your Turn!

1. Factor $12x^2 - 16x - 3$ by grouping.
   $(6x + 1)(2x - 3)$

*Encourage students to use a variety of methods that fit their strengths as natural scaffolding for attempting to factor by grouping.*

*Students who struggle with factoring by grouping may be able to guess and check efficiently to find the factors. Prompt those students to find a partner who did factor by grouping and make connections between their representation of grouping and the factors they found.*

# 7.6.6 Wrap-Up: Ticket Out the Door

***Component Overview**: Students will complete a Ticket Out the Door that assesses their knowledge of factoring. This activity should be done independently to provide data on student understanding.*

## 7.6.6 Wrap-Up: Ticket Out the Door

1. Factor each expression.

   a. $h^2 - 12h + 27$
   $(h-9)(h-3)$

   b. $36a^4b^7 + 42a^6b^3$
   $6a^4b^3(6b^4 + 7a^2)$

   c. $12x^2 - 8x - 15$
   $(6x+5)(2x-3)$

   d. $5x^2 + 37x + 14$
   $(5x+2)(x+7)$

*This Ticket Out the Door should be used to help determine students who need support on each type of factoring. This can serve as a "checkpoint" for the content covered thus far in this unit.*

*If time is a concern, consider assigning the 7.6.5 Your Turn! for homework to ensure adequate time for students to complete this Wrap-Up.*

# 7.7 Factoring Perfect Square Trinomials

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system. | Learning Target | • I can factor perfect square trinomials. |

| Benchmark Coherence | Building On | Working Toward |
|---|---|---|
| | MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions. | MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system. |

| | |
|---|---|
| Essential Questions | • How can you determine if a trinomial is a perfect square trinomial?<br>• How can recognizing that a polynomial is a perfect square trinomial help with factoring? |
| Lesson Narrative | In this lesson, students will be introduced to the concept of a perfect square trinomial, learning what they are and how to identify them. They will then transition into learning how to factor a perfect square trinomial before practicing identifying and factoring perfect square trinomials. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 7.7.1 Warm-Up (~5 min.) | Create equivalent forms of expressions *(SE p. 391)* | 7.7.4 Guided Practice (10 min.) | Guided practice with factoring perfect square trinomials *(SE p. 396)* |
| | 7.7.2 Collaboration (10-15 min.) | Use algebra tiles to explore perfect square trinomials *(SE p. 392)* | 7.7.5 Your Turn! (5 min.) | Independent practice with factoring perfect square trinomials *(SE p. 396)* |
| | 7.7.3 Guided Instruction (10 min.) | Formal instruction on factoring perfect square trinomials *(SE p. 395)* | 7.7.6 Wrap-Up (~5 min.) | Emoji self-reflection activity *(SE p. 397)* |

| Resources | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| | <u>Key Terms:</u><br>• Perfect square trinomial<br><br><u>Additional Terms:</u> N/A | • (Optional) Algebra tiles | • For 7.7.4 Guided Practice, consider creating an index card with the answers or a printout of the table to quickly identify where students are making errors during the practice while circulating the room. |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may believe any larger numbers or coefficients cannot be perfect square trinomials (like question 1 in Your Turn!).</li><li>Students may mistake a difference of squares for a perfect square trinomial.</li></ul></td><td><ul><li>Students should not see perfect square trinomials as something else or new or an addition to what they’ve learned. The purpose is for students to discover the relationship between a perfect square trinomial and its factored form. Students can still use other methods for factoring these expressions.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect<br>7.7.3 Guided Instruction provides students with guided instruction on factoring perfect square trinomials. During this activity, ensure that students can hear your thinking aloud during instruction. This way, students can connect this thinking to the content from start to finish.</td><td>MA.K12.MTR.4.1: Discussion and Reflection<br>7.7.2 Collaboration provides students with an activity where they work with a partner to explore perfect square trinomials using algebra tiles. Students will communicate mathematical ideas, vocabulary, and methods effectively. Consider ways to establish a culture in which students ask questions of the teacher and their peers and error is seen as an opportunity for learning.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 7.7.2 Collaboration, students will work together. Consider grouping students based on their learning styles. This can help with the way students connect to the concept of perfect square trinomials (i.e., with physical algebra tiles). Consider providing physical algebra tiles or a paper cut-out version in order to provide a physical manipulative for the algebra tile portions of the activity.<br><br>7.7.5 Your Turn! provides students with an independent practice opportunity with factoring perfect square trinomials. Ensure that students show their work so you can see if they recognize that they are perfect square trinomials or if they are factoring as a standard trinomial.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.7.1 Warm-Up: Critical Thinking

***Component Overview**: Students will complete a table with equivalent quadratic expressions in standard and factored form.*

## 7.7.1 Warm-Up: Critical Thinking

1. Fill in the blanks with an integer so that the quadratic expression is factorable. Then, complete the factored form.
   *Answers will vary.*

| Quadratic Expression | Factored Form |
|---|---|
| $x^2 + 12x + 36$ | $(x + 6)(x + 6)$ |
| $x^2 - 9x - 36$ | $(x - 12)(x + 3)$ |
| $x^2 + 9x - 36$ | $(x - 3)(x + 12)$ |

*Use the first expression as a launch into perfect square trinomials.*

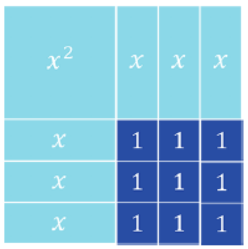
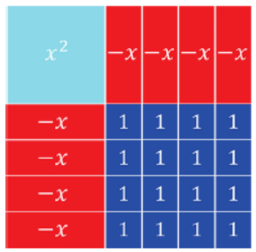
- *How is the first expression different than the rest?*
- *Is there a way to efficiently identify when that may happen?*
- *What does that make you think about factoring?*

# 7.7.2 Collaboration: Perfect Square Trinomials

***Component Overview***: *Students will work collaboratively to discover the concept of perfect square trinomials. This discovery will be done using algebra tiles. For this activity, students should work in pairs or small groups to complete the activity.*

## 7.7.2 Collaboration: Perfect Square Trinomials

1. The quadratic expressions represented by algebra tiles are perfect square trinomials.

| A | B |
|---|---|
|  |  |

a. Discuss with your partner why these quadratic expressions are called perfect square trinomials.
*Answers will vary. The tiles are laid out in a square.*

b. Write the trinomials represented by the algebra tiles.

| Trinomial A | Trinomial B |
|---|---|
| $x^2 + 6x + 9$ | $x^2 - 8x + 16$ |

c. Use the algebra tile representations to write the factored form of each trinomial.

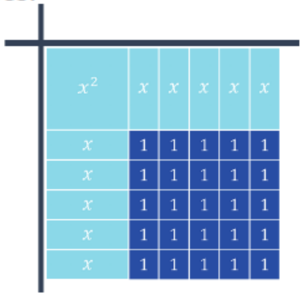
| Factored Form of Trinomial A | Factored Form of Trinomial B |
|---|---|
| $(x + 3)(x + 3)$ | $(x - 4)(x - 4)$ |

*For this activity, release students to work with a partner or small group to complete each part. For part a, this is a discussion question with less focus on writing their entire description in the book. When reviewing the answers, have multiple partner pairs or groups share their responses and rationales.*

*How did you determine the factored form of Trinomials A and B?*

# 7.7.2 Collaboration: Perfect Square Trinomials (continued)

2. The quadratic expression $x^2 + 10x + 25$ is represented by the algebra tiles.



a. Pranjal used the algebra tiles to write the factored form of $x^2 + 10x + 25$ as $(x+5)^2$.

Check Pranjal's work by completing the steps.

$$(x+5)^2 = (x+5)(x+5)$$

$$(x+5)(x+5) = (x+5)^2$$

b. Discuss with your partner how writing the factored form of the quadratic expression $x^2 + 10x + 25$ as $(x+5)^2$ relates to the expression being called a perfect square trinomial. Summarize your discussion.

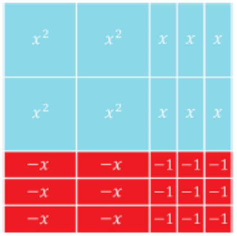
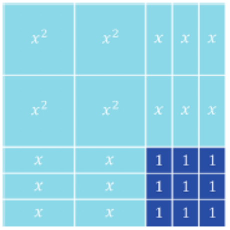
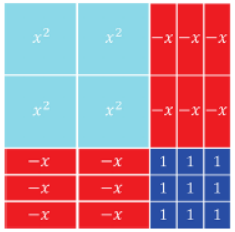
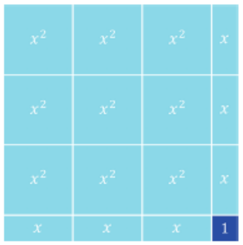
*Answers will vary. The binomial is squared.*

*If available, consider providing students with physical or virtual algebra tiles. This can help learners have a hands-on entry point into this question and others to explore the information provided in this activity.*

*Agree or Disagree*
*Highlight student responses that showcase conceptual understanding by pausing for a poll to agree or disagree, focusing on the rationale behind your decision rather than the decision itself.*

# 7.7.2 Collaboration: Perfect Square Trinomials (continued)

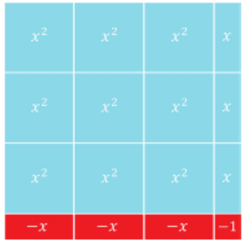
3. Work with your partner to determine which quadratic expressions are perfect square trinomials.

| Expression | Algebra Tiles |
|---|---|
| $4x^2 - 9$<br><br>Is it a perfect square trinomial?<br>○ Yes<br>○ No |  |
| $4x^2 + 12x + 9$<br><br>Is it a perfect square trinomial?<br>○ Yes<br>○ No |  |
| $4x^2 - 12x + 9$<br><br>Is it a perfect square trinomial?<br>○ Yes<br>○ No |  |
| $9x^2 + 6x + 1$<br><br>Is it a perfect square trinomial?<br>○ Yes<br>○ No |  |

*What about this expression makes it not a perfect square trinomial?*

*Highlight the difference between the difference of squares (no need to use this vocabulary yet, as this will be introduced in Lesson 8) and perfect square trinomials. Some students may be confused on the first one because there are no obvious “holes” in the mat. It appears to be a square even though its length and width are not the same.*

# 7.7.2 Collaboration: Perfect Square Trinomials (continued)

| | |
|---|---|
| $9x^2 - 1$<br><br>Is it a perfect square trinomial?<br>○ Yes<br>○ No |  |

*How can you tell, by looking at the algebra tiles, that this is not a perfect square trinomial?*

4. Describe some key characteristics of perfect square trinomials by relating both the representation of a quadratic expression using algebra tiles and the trinomial.
   *Answers will vary. A perfect square trinomial will create a square when representing it using algebra tiles. The $a$-term and the $c$-term are perfect squares.*

*When reviewing answers, use clear and concise language to highlight how the algebra tiles show that a perfect square trinomial exists.*

# 7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

***Component Overview***: *Students will receive formal instruction on factoring perfect square trinomials. They will be introduced to the definition of a perfect square trinomial and the process for factoring them.*

## 7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

A ***perfect square trinomial*** is a trinomial of the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$.

1. The quadratic expression $4x^2 + 12x + 9$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.
   $(2x)^2 + 2(2x)(3) + 3^2$

Perfect square trinomials of the form $a^2 + 2ab + b^2$ can be factored as $(a+b)^2$.

2. Factor $4x^2 + 12x + 9$.
   $(2x+3)^2$

3. The quadratic expression $9x^2 - 12x + 4$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.
   $(3x)^2 - 2(3x)(2) + 2^2$

Perfect square trinomials of the form $a^2 - 2ab + b^2$ can be factored as $(a-b)^2$.

4. Factor $9x^2 - 12x + 4$.
   $(3x-2)^2$

*Add the definition of a perfect square trinomial to any existing anchor charts, Frayer Models, or guided notes that students have for the factoring processes learned in this unit.*

*Students have intentionally already explored perfect square trinomials. They now have prior knowledge to activate during this activity rather than approaching instruction as something completely new. Use inquiry to activate connections from the Collaboration activity.*

- $4x^2$ *is a perfect square. What is the square root of* $4x^2$*? This is the value of* $a$.
- *Which other term in the given expression is a perfect square? What is its square root? This is the value of* $b$.
- *What is a way to determine if a trinomial would be classified as a perfect square trinomial?*
- *What is the benefit to the factoring process by knowing that a trinomial is a perfect square trinomial?*

# 7.7.4 Guided Practice: Factoring Perfect Square Trinomials

***Component Overview***: *Students will practice factoring perfect squares by identifying, given a quadratic expression, whether it meets the conditions of a perfect square trinomial.*

## 7.7.4 Guided Practice: Factoring Perfect Square Trinomials

1. Place a check mark in the appropriate columns of the table to indicate whether or not each quadratic expression is a perfect square trinomial. If it is a perfect square, write its factored form as $(a+b)^2$ or $(a-b)^2$.

| Quadratic Expression | First Term is a Perfect Square | Third Term is a Perfect Square | Second Term is $2ab$ or $-2ab$ | Factored Form |
|---|---|---|---|---|
| $2g^2+6g+9$ | | ✓ | | *Prime* |
| $16y^2-56y+49$ | ✓ | ✓ | ✓ | $(4y-7)^2$ |
| $b^2-26b+25$ | ✓ | ✓ | | $(b-25)(b-1)$ |
| $4w^2-16w+8$ | ✓ | | | $4(w^2-4w+2)$ |
| $9x^2-4$ | ✓ | ✓ | | $(3x+2)(3x-2)$ |
| $q^2-2qz+z^2$ | ✓ | ✓ | ✓ | $(q-z)^2$ |

*Consider creating a small version of the table answer key or printing a small version to keep handy. This will be helpful when quickly scanning student work to see where potential errors are being made to intervene more quickly.*

# 7.7.5 Your Turn!

***Component Overview***: *Students will work independently to factor two perfect square trinomials.*

## 7.7.5 Your Turn!

1. Factor $16h^2 + 88h + 121$.
   $(4h + 11)^2$

2. Factor $k^2 - 10k + 25$.
   $(k - 5)^2$

*For this activity, allow students to try these on their own to determine their level of mastery of how to factor these types of trinomials.*

# 7.7.6 Wrap-Up: Reflection

***Component Overview**: Students will complete an emoji-scale self-reflection activity on their understanding of perfect square trinomials.*

## 7.7.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of factoring perfect square trinomials.
   *Answers will vary.*


2. Complete the statement.
   *Answers will vary.*

I chose this emoji because ______________________________

_____________________________________________________

_____________________________________________________

____________________________________________________.

*Prompt students to pair their responses with an example that demonstrates their level of understanding. Students who mark 1-5 on the scale should use Mark-Up text to identify at which point they have questions, while students who mark 6-10 can show an example or find a partner who marked 1-5 and assist.*

# 7.8 Factoring the Difference of Two Squares

<table>
<tr><th colspan="6">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.</td><td>Learning Targets</td><td colspan="2"><ul><li>I can factor a difference of two squares.</li><li>I can rewrite a polynomial expression as a product of polynomials by factoring.</li></ul></td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="3">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="3">MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.</td><td colspan="2">MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.</td></tr>
<tr><td>Essential Questions</td><td colspan="5"><ul><li>How can a polynomial with two terms be factored?</li><li>How can you recognize and factor a difference of two squares?</li></ul></td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this lesson, students conclude their journey with factoring by learning about the final special case included in this unit, a difference of two squares. Students begin by exploring a difference of squares using algebra tiles. They will then practice identifying whether a given expression is a difference of squares. Students will then learn how to factor a situation represented by a difference of squares. The lesson includes components of mixed-review practice of factoring given a variety of polynomials.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>7.8.1 Warm-Up<br>(~5 min.)</td><td>Factor perfect square trinomials (SE p. 398)</td><td>7.8.4 Try It<br>(10 min.)</td><td colspan="2">Mixed practice factoring polynomials (SE p. 401)</td></tr>
<tr><td>7.8.2 Exploration<br>(15 min.)</td><td>Exploration activity to discover a difference of two squares (SE p. 398)</td><td>7.8.5 Your Turn!<br>(5-10 min.)</td><td colspan="2">Independent practice factoring polynomials (SE p. 402)</td></tr>
<tr><td>7.8.3 Guided Practice<br>(10 min.)</td><td>Guided practice with factoring difference of two squares (SE p. 400)</td><td>7.8.6 Wrap-Up<br>(~5 min.)</td><td colspan="2">Self-reflection on student understanding of factoring from this unit (SE p. 402)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td colspan="3">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u><ul><li>Difference of two squares</li></ul><u>Additional Terms:</u> N/A</td><td><ul><li>N/A</li></ul></td><td colspan="3"><ul><li>N/A</li></ul></td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may mistake the difference of squares for a perfect square trinomial.</li><li>Students may not recognize a difference of squares where the leading coefficient is negative but the sign between the terms is addition.</li></ul></td><td><ul><li>A difference of squares was shown pictorially in Lesson 7 but not formally introduced. This lesson provides not only instruction on difference of squares but also a mixed review of factoring all polynomial types explored in this unit.</li><li>If time is an issue, consider assigning 7.8.5 Your Turn! for homework.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Poll the Class<br>The table component in 7.8.2 Exploration provides an opportunity for students to determine if a series of conditions are met for a given polynomial. Consider leveraging this protocol when reviewing the answers to the table to determine who in the class thinks it meets a given condition and who does not.</td><td>MA.K12.MTR.3.1: Mathematical Fluency<br>7.8.4 Try It provides students with a mixed-review opportunity for factoring polynomials using the methods learned throughout this unit. Students will select efficient and appropriate methods for factoring given expressions. They should maintain flexibility and accuracy when factoring different types of polynomials. The lesson offers multiple opportunities for students to practice efficient and generalizable methods.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">When reviewing 7.8.2 Exploration, have students add the formal definition, with examples and nonexamples, to their current factoring flip charts, Frayer Models, or guided notes that they have for this unit.<br><br>For 7.8.4 Try It, consider grouping students together based on strengths from this unit. This activity is a mixed review, and having students grouped by strengths from the unit can help them learn better from one another.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

# 7.8.1 Warm-Up: Bell Work

***Component Overview**: Students will complete two factoring problems that are both perfect square trinomials.*

### 7.8.1 Warm-Up: Bell Work

1. Factor $36x^2 - 12x + 1$.
   $(6x - 1)^2$

2. Factor $9y^2 + 30y + 25$.
   $(3y + 5)^2$

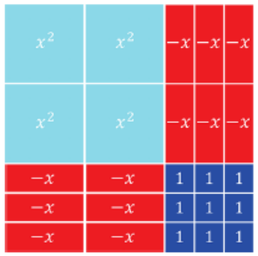
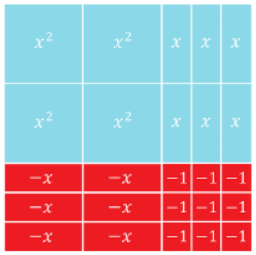
*Use inquiry to extend student thinking from Lesson 7. Is this a perfect square trinomial? How do you know? How did that impact your method for factoring?*

# 7.8.2 Exploration: What Is a Difference of Two Squares?

***Component Overview***: *Students will work together to explore the concept of a difference of two squares. They will use algebra tiles to conceptually develop understanding of this concept and relate it to what they've learned in this unit.*

## 7.8.2 Exploration: What is a Difference of Two Squares?

The table shows two algebra tile models and their corresponding expressions.

| $4x^2 - 12x + 9$ | $4x^2 - 9$ |
|---|---|
|  |  |

The quadratic expression $4x^2 - 12x + 9$ is a perfect square trinomial.

1. Discuss with your partner why the quadratic expression $4x^2 - 9$ is not a perfect square trinomial, even though the algebra tile model is a square. Summarize your discussion.
   *There is no middle term to meet the conditions of a perfect square trinomial.*

2. Use the algebra tile models to factor each quadratic expression.

| $4x^2 - 12x + 9$ | $4x^2 - 9$ |
|---|---|
| $(2x-3)^2$ *or* $(2x-3)(2x-3)$ | $(2x+3)(2x-3)$ |

3. What do you notice about the two factored forms?
   *They are similar in terms, but the second cannot be written as a square since there are two binomials with different signs. Or: In the first, the signs are the same for both binomials, and in the second, the signs are different.*

*This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.*

*Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.8.3 Guided Practice.*

*Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.*

*Monitor for student responses to revisit or highlight during whole group synthesis. For example, students may identify the similarities of using square roots while also highlighting the difference in a missing $b$ term. Prompt this response whole group later as a launch for synthesizing the difference between a perfect square trinomial and a difference of two squares.*

4. The quadratic expression $4x^2 - 9$ is a difference of two squares. Discuss with your partner why $4x^2 - 9$ is called a "difference."
*Answers will vary. The difference refers to the subtraction.*

5. The two squares shown represent each term of the binomial $4x^2 - 9$.

| $4x^2$ | $9$ |
|---|---|

What is the side length of each square?

| $4x^2$ | $9$ |
|---|---|
| Side Length = $\underline{2x}$ | Side Length = $\underline{3}$ |

The squares are a geometric representation of each term. The quadratic expression $4x^2 - 9$ can be rewritten with each term as a monomial squared, as shown.

$$(2x)^2 - (3)^2$$

6. Discuss with your partner how the factored form of $4x^2 - 9$ reflects the expression $(2x)^2 - (3)^2$. Summarize your discussion.
*Answers will vary. The first term of each binomial is $2x$ and the second term is $3$.*

The **difference of two squares** is a binomial of the form $(a)^2 - (b)^2$, where $a$ and $b$ can represent a monomial expression or any value except $0$. The factored form is $(a + b)(a - b)$.

*Provide an opportunity for students to share their answers to questions 5 and 6.*

*Have students add the formal definition of the difference of two squares into their existing guided notes, anchor charts, flip charts, foldables, Frayer Models, or other reference documents created for this unit.*

# 7.8.2 Exploration: What Is a Difference of Two Squares? (continued)

7. Use the table to determine if each expression is a difference of two squares.

| Quadratic Expression | The Expression is a Binomial that Can Be Written as a Difference | First Term is a Perfect Square | Second Term is a Perfect Square |
|---|---|---|---|
| $-100g^4 + 1$ | o Yes o No | o Yes o No | o Yes o No |
| $16y^2 - 100y$ | o Yes o No | o Yes o No | o Yes o No |
| $-b^2 + 25$ | o Yes o No | o Yes o No | o Yes o No |
| $25w^2 - 40w + 16$ | o Yes o No | o Yes o No | o Yes o No |
| $9x^2 - 4y^6$ | o Yes o No | o Yes o No | o Yes o No |
| $q^2 - 2qz + z^2$ | o Yes o No | o Yes o No | o Yes o No |
| $49g^2 - 81h^2$ | o Yes o No | o Yes o No | o Yes o No |

a. Complete the table.

b. Circle the expressions in the table that are considered a difference of two squares, based on your check marks.

*This table purposely starts with a binomial that is a difference of two squares but is not in the recognizable form. The 1 is also purposely used to ensure that you are able to talk about leading terms that are numeric. This is also seen again in the third row.*

*Allow students to work through question 7 with their partners, but be sure to review any errors, paying particular attention to these examples, as needed before transitioning to the next component.*

# 7.8.3 Guided Practice: Factoring a Difference of Two Squares

***Component Overview***: *Students will practice factoring the difference of squares with three different polynomials. For this activity, students will need a partner in order to engage in all the questions.*

## 7.8.3 Guided Practice: Factoring a Difference of Two Squares

*Use observational data from the 7.8.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in the 7.8.2 Exploration before releasing students to work on questions 2 and 3 with their partners.*

1. Consider the expression $49g^2 - 81h^2$.

   a. Rewrite $49g^2 - 81h^2$ as a difference of two squares.
   $(7g)^2 - (9h)^2$

   b. Factor $49g^2 - 81h^2$.
   $(7g + 9h)(7g - 9h)$

2. Consider the expression $9x^2 - 4y^6$.

   a. Rewrite $9x^2 - 4y^6$ as a difference of two squares.
   $(3x)^2 - (2y^3)^2$

   b. Factor $9x^2 - 4y^6$.
   $(3x + 2y^3)(3x - 2y^3)$

3. Both $-121y^2 + 81$ and $-1 + 16x^2$ are differences of two squares. Discuss with your partner how to rewrite them as a difference of two squares. Then, work together to factor each.

| $-121y^2 + 81$ | $-1 + 16x^2$ |
|---|---|
| $81 - 121y^2$<br><br>$(9 - 11y)(9 + 11y)$ | $16x^2 - 1$<br><br>$(4x + 1)(4x - 1)$ |

# 7.8.4 Try It: Factoring Polynomials

***Component Overview**: Students will practice factoring using the methods learned throughout the unit.*

## 7.8.4 Try It: Factoring Polynomials

This is a mixed practice, which includes all of the different types of polynomials factored in this unit. If necessary, review previous lessons for examples of how to factor each type.

*This activity is not limited to polynomials that can be factored using the difference of squares. Additional methods from prior lessons will be necessary to completely factor each expression.*

1. Factor each expression.

   a. $49t^2 - 42t + 9$
   $(7t - 3)^2$

   b. $1 - 25k^2$
   $(1 + 5k)(1 - 5k)$

   c. $36x^2y^7 - 48x^4y^4 + 12x^2y^4$
   $12x^2y^4(3y^3 - 4x^2 + 1)$

   d. $12k^2 + 3k - 15$
   $3(k - 1)(4k + 5)$

   e. $d^2 + d - 42$
   $(d - 6)(d + 7)$

*Consider having students write the methods that they used to factor each of the expressions in this component. Provide students a reference point to specific lessons they can go back to if they get stumped.*

# 7.8.5 Your Turn!

***Component Overview***: *Students will independently practice factoring polynomials using different methods learned throughout the unit.*

## 7.8.5 Your Turn!

1. Factor each expression.

a. $25m^2 + 20m + 4$
$(5m + 2)^2$

b. $2g^2 + 3g - 20$
$(2g - 5)(g + 4)$

c. $y^2 - 5y - 84$
$(y - 12)(y + 7)$

d. $49z^4 - 16w^4$
$(7z^2 + 4w^2)(7z^2 - 4w^2)$

*This activity is not limited to polynomials that can be factored using the difference of squares. Additional methods from prior lessons will be necessary in order to completely factor each one.*

*Consider having students write the methods that they used to factor each of the questions in this activity. Provide students a reference point to specific lessons they can go back to if they get stumped.*

# 7.8.6 Wrap-Up: Reflection

***Component Overview**: Students will complete a self-reflection on their levels of understanding of each of the factoring methods learned throughout the unit.*

## 7.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.
   *Answers will vary.*

   I need help. I got it. I could help someone.

| | |
|---|---|
| Factoring out a common monomial factor | |
| Factoring a trinomial where $a = 1$ | |
| Factoring a trinomial where $a > 1$ | |
| Factoring a perfect square trinomial | |
| Factoring a difference of two squares | |

*What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of these learning targets throughout the unit and how to best address potential gaps in understanding prior to the unit assessment.*

*Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.*

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